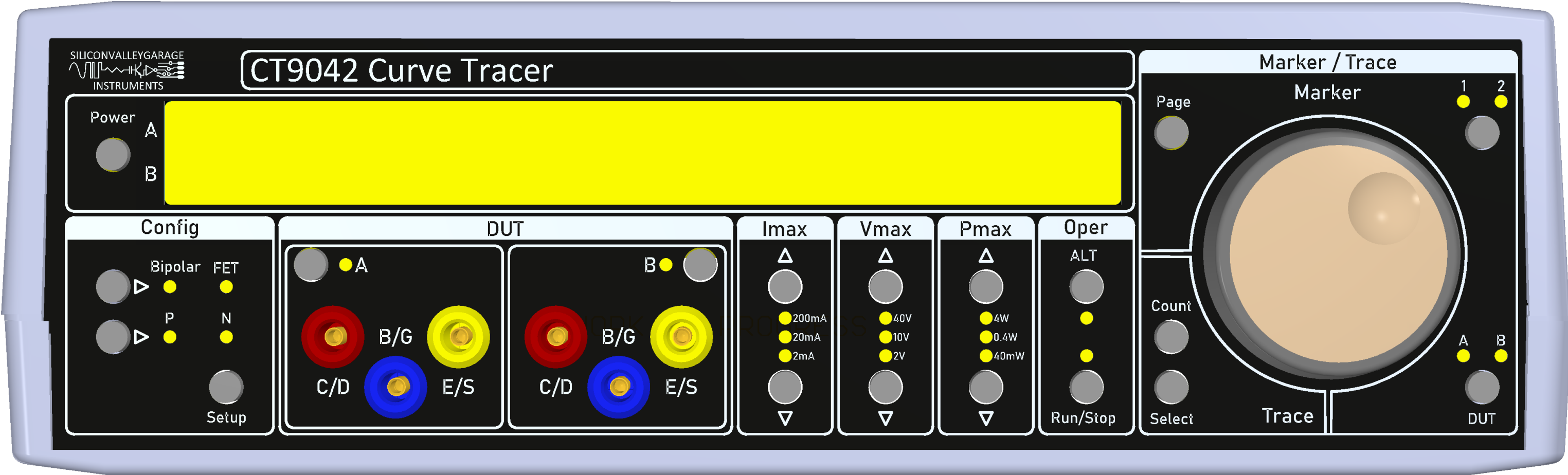


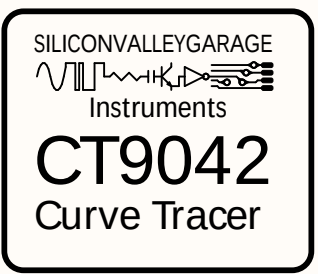
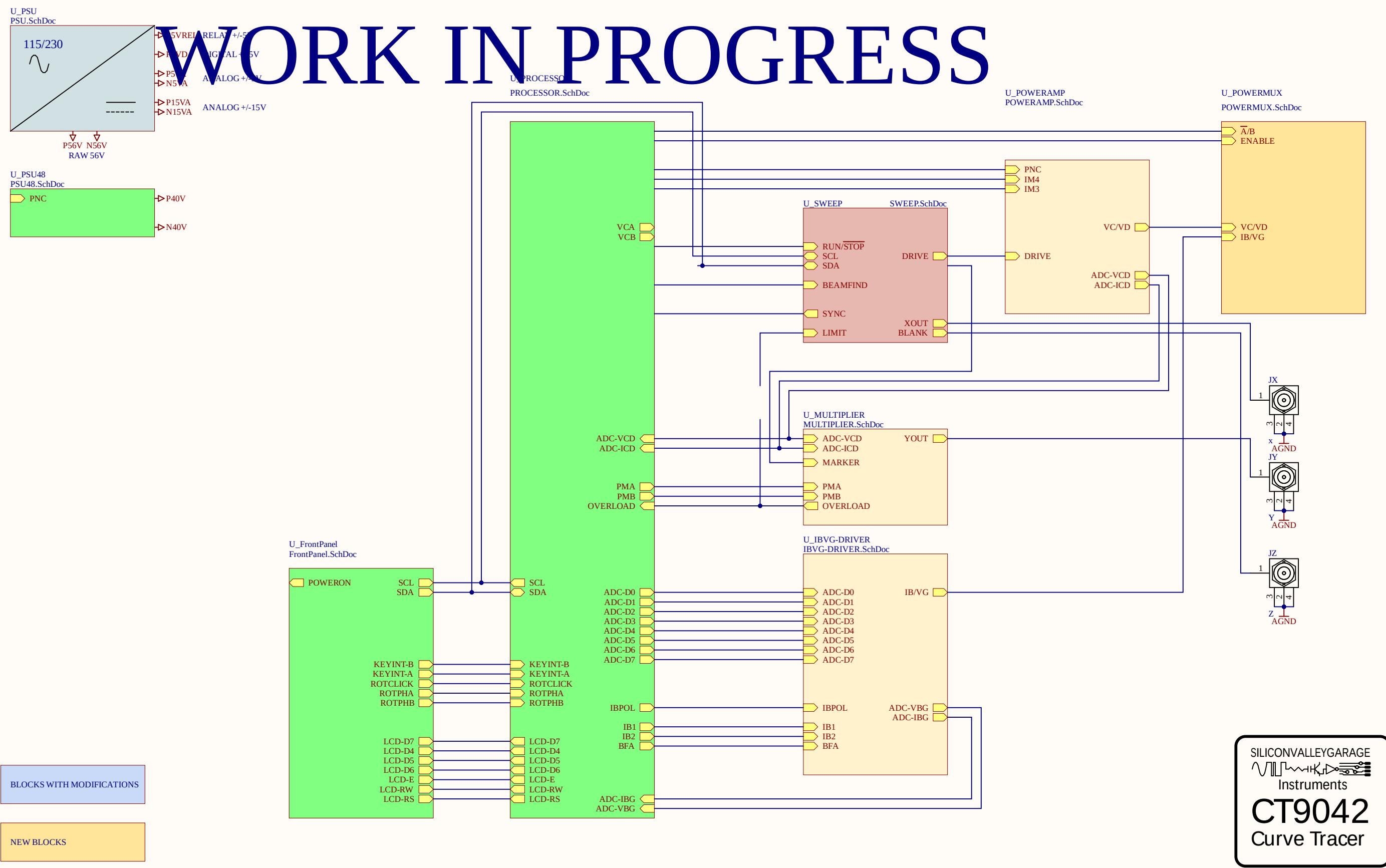
Realistic View



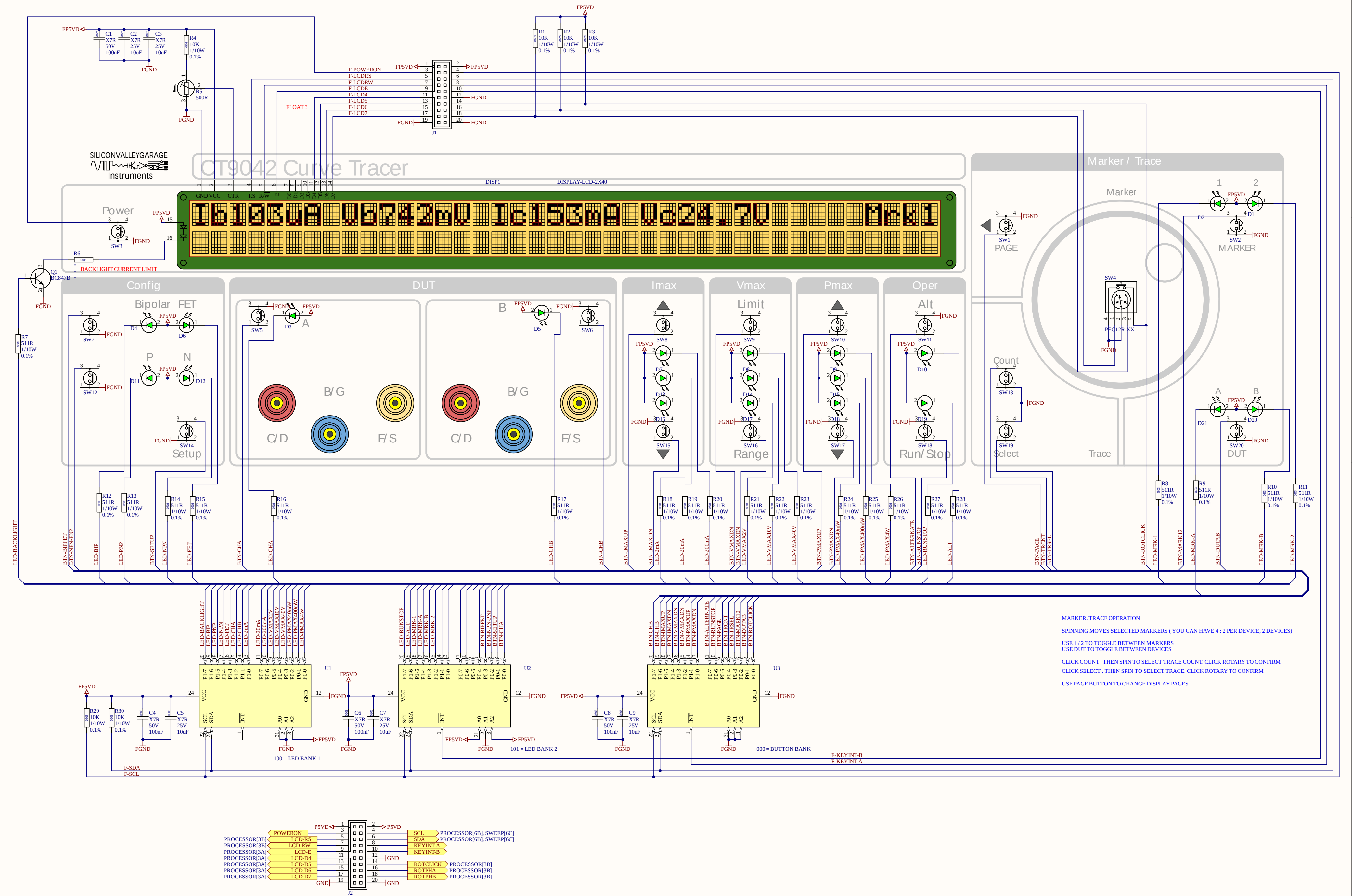
Document Creation Date: 9/6/2025

Design : free_electron

WORK IN PROGRESS



CT9042 Curve Tracer



MARKER/TRACE OPERATION

SPINNING MOVES SELECTED MARKERS (YOU CAN HAVE 4 : 2 PER DEVICE, 2 DEVICES)

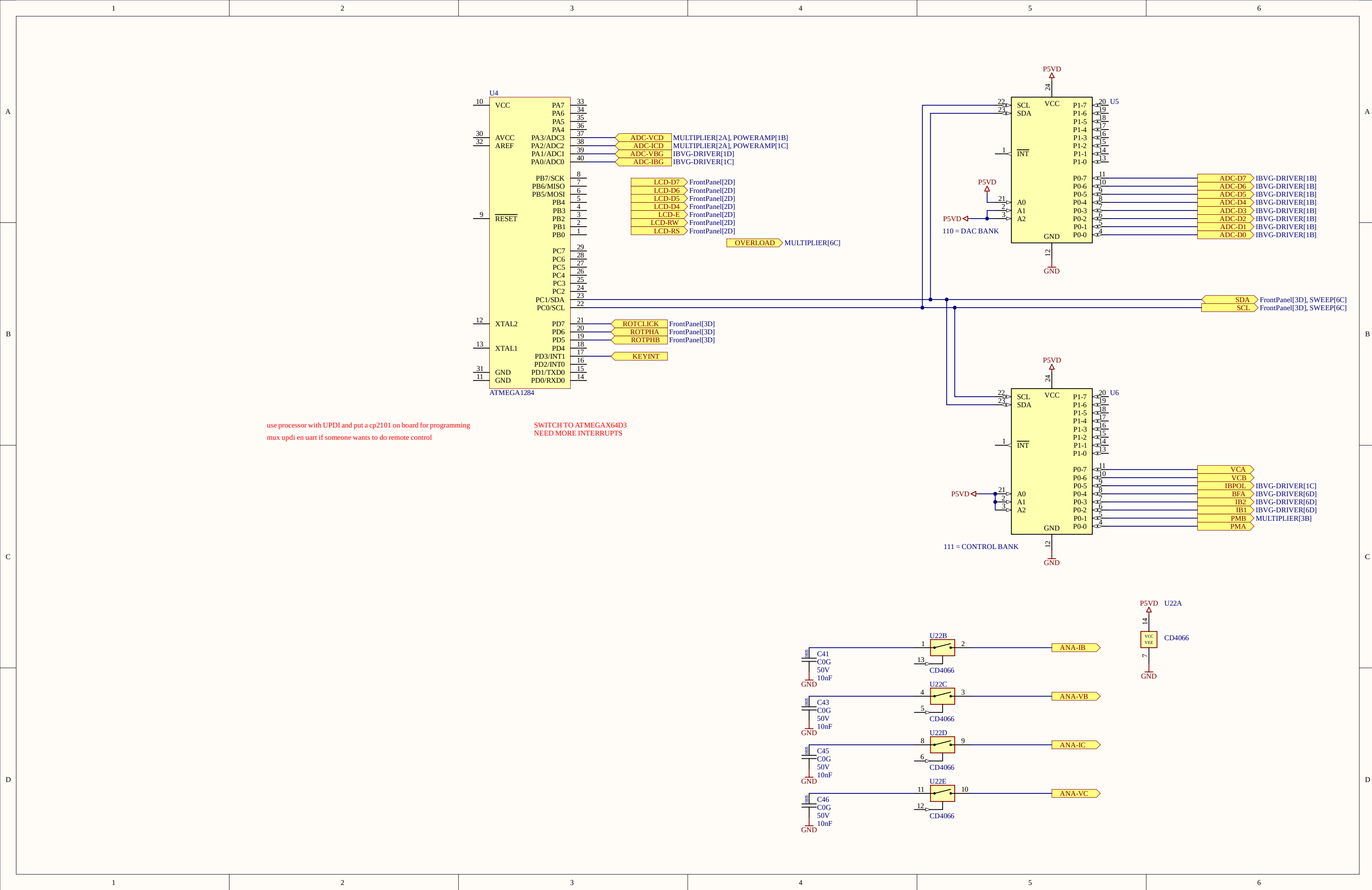
USE 1 / 2 TO TOGGLE BETWEEN MARKERS

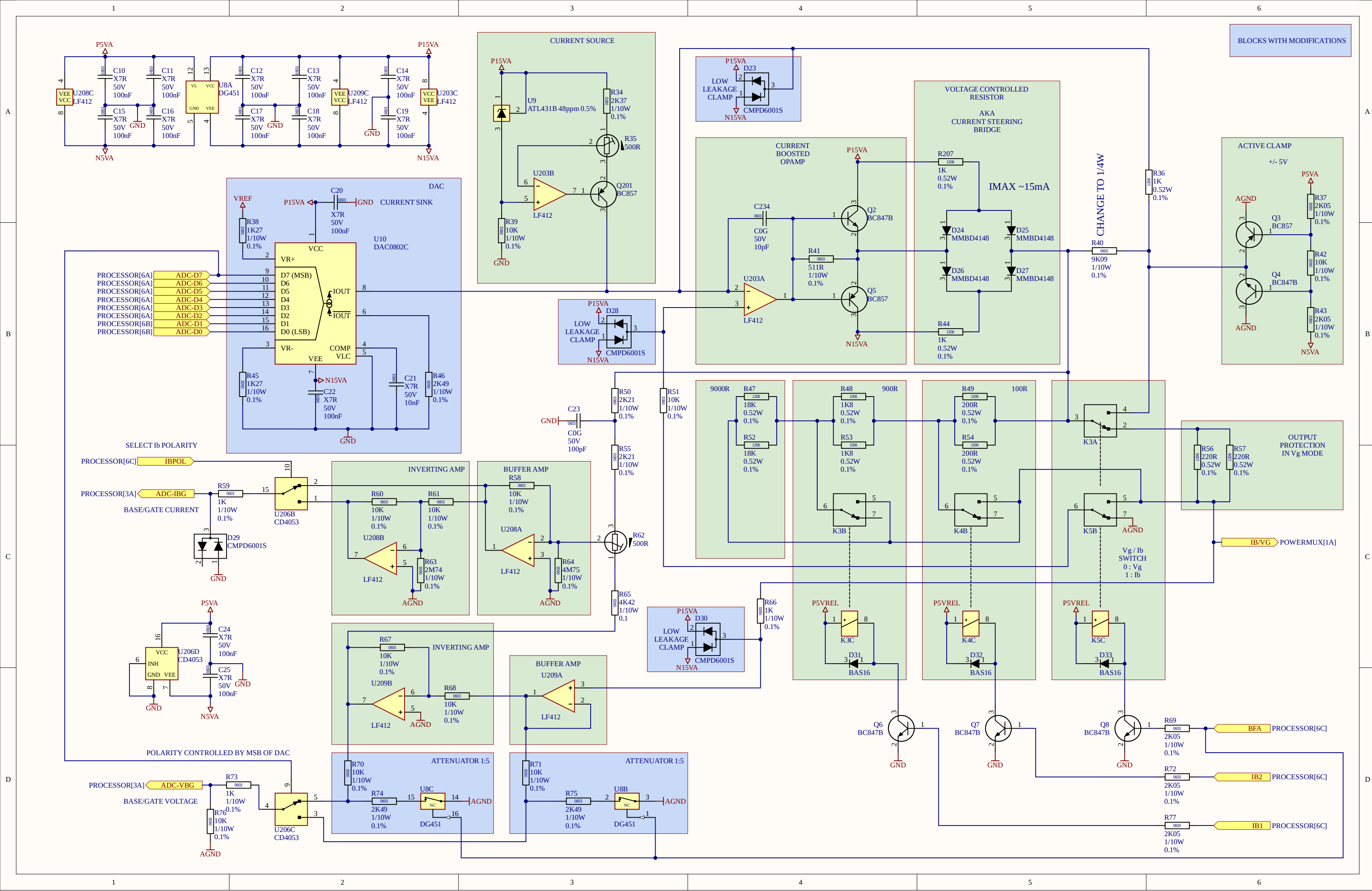
USE DUT TO TOGGLE BETWEEN DEVICES

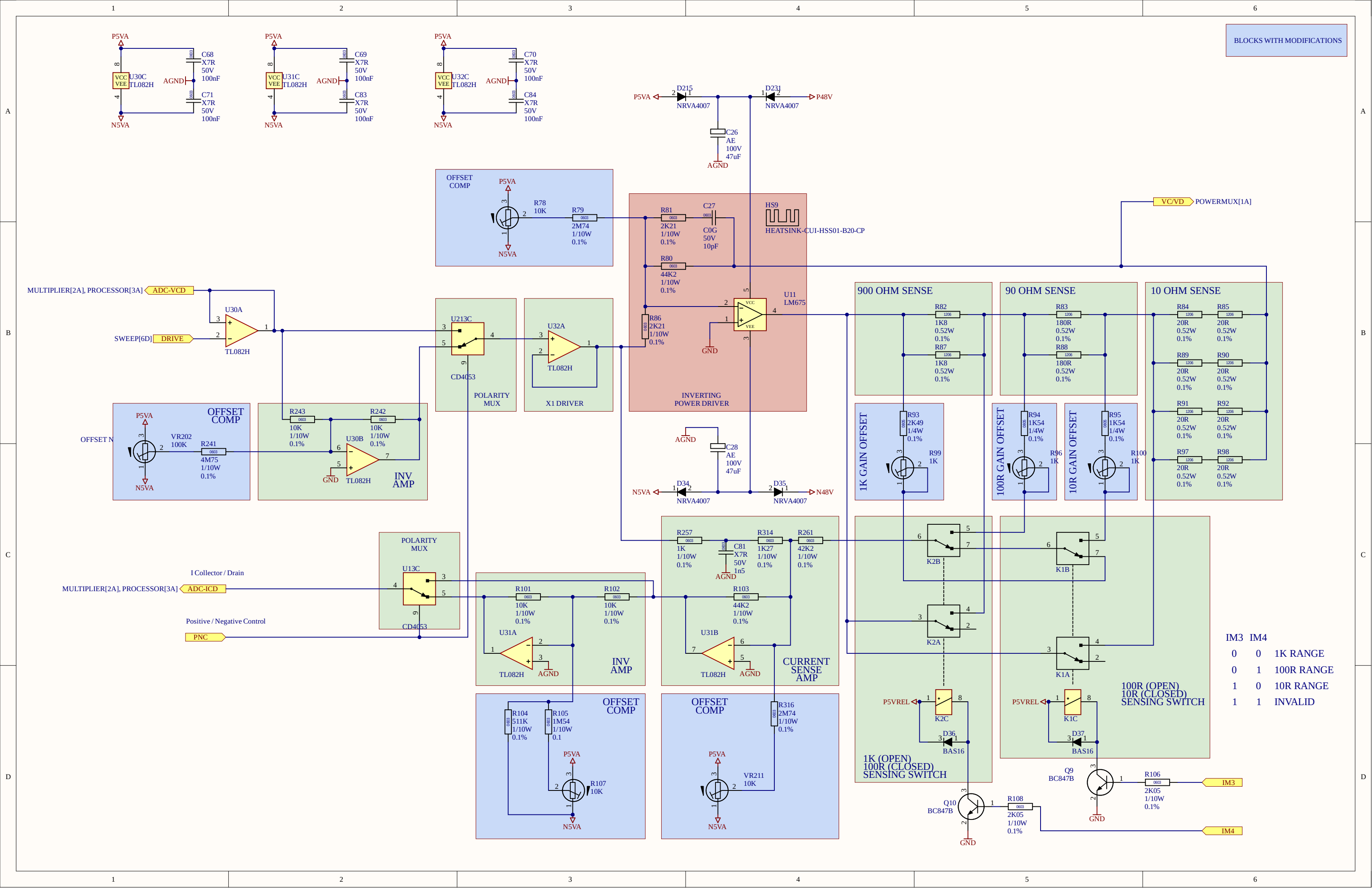
CLICK COUNT, THEN SPIN TO SELECT TRACE COUNT. CLICK ROTARY TO CONFIRM

CLICK SELECT, THEN SPIN TO SELECT TRACE. CLICK ROTARY TO CONFIRM

USE PAGE BUTTON TO CHANGE DISPLAY PAGES

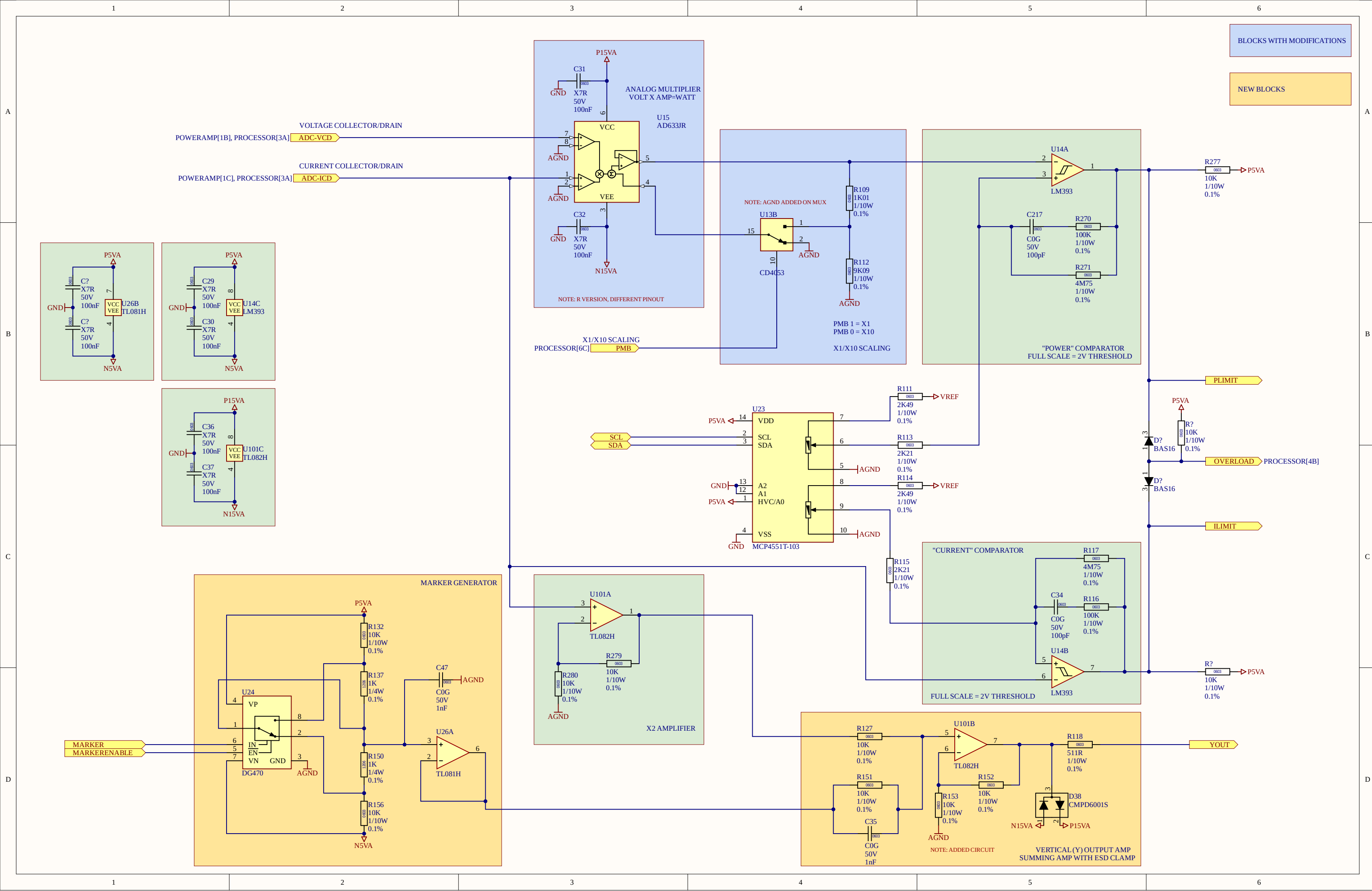


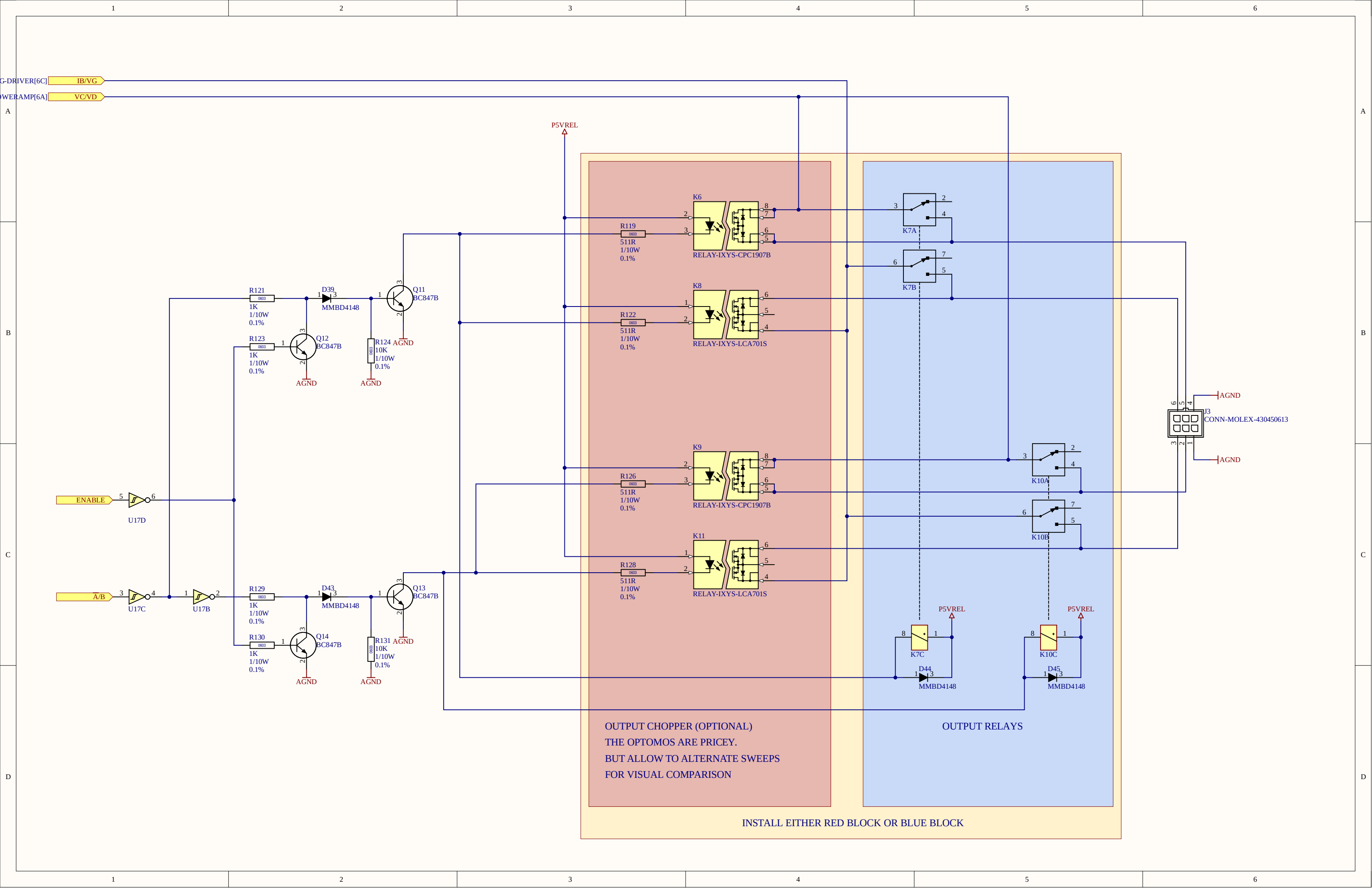


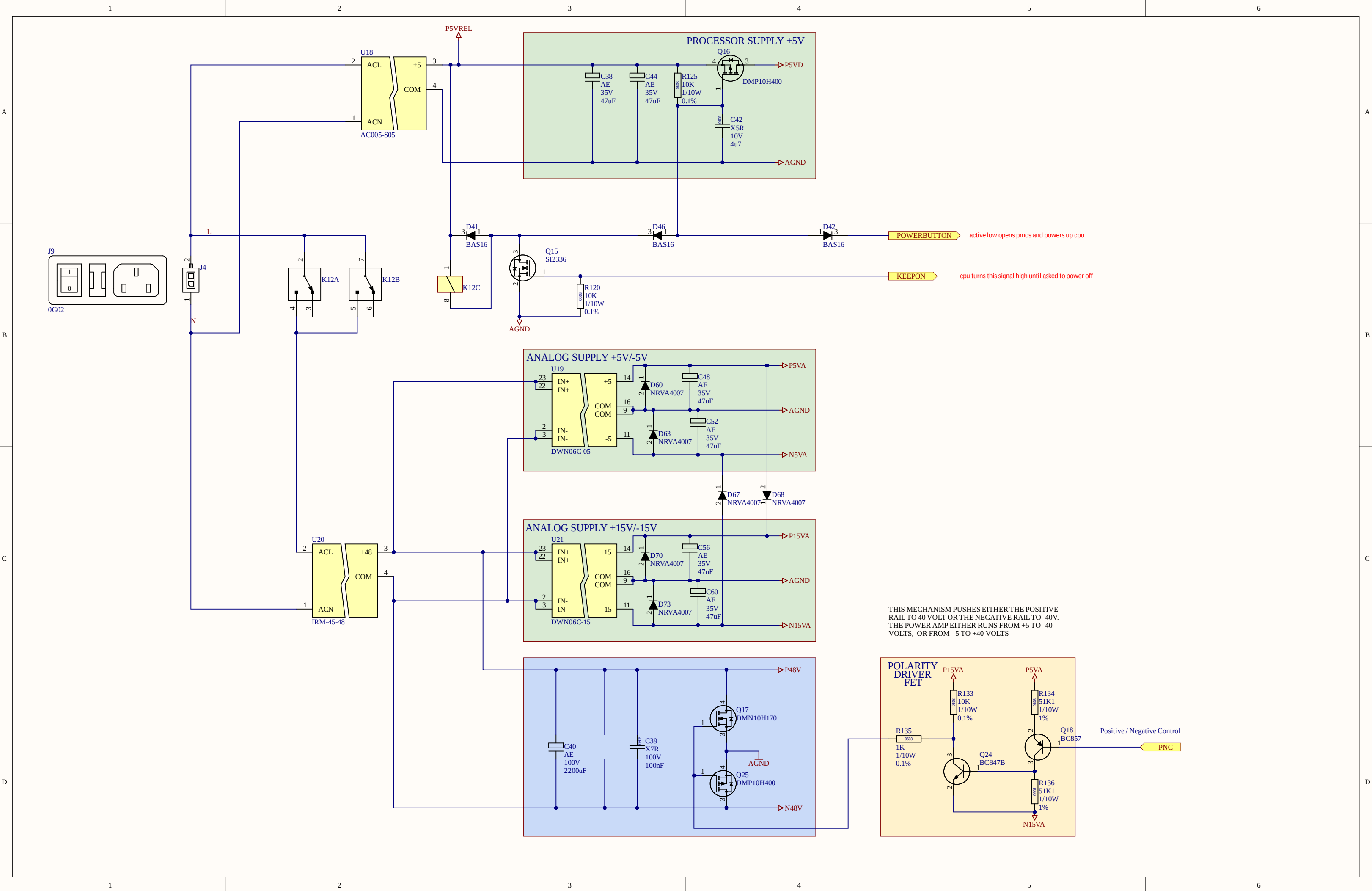


BLOCKS WITH MODIFICATIONS

		IM3	IM4	
0	0	1K RANGE		
0	1	100R RANGE		
1	0	10R RANGE		
1	1	INVALID		







A

A

B

B

C

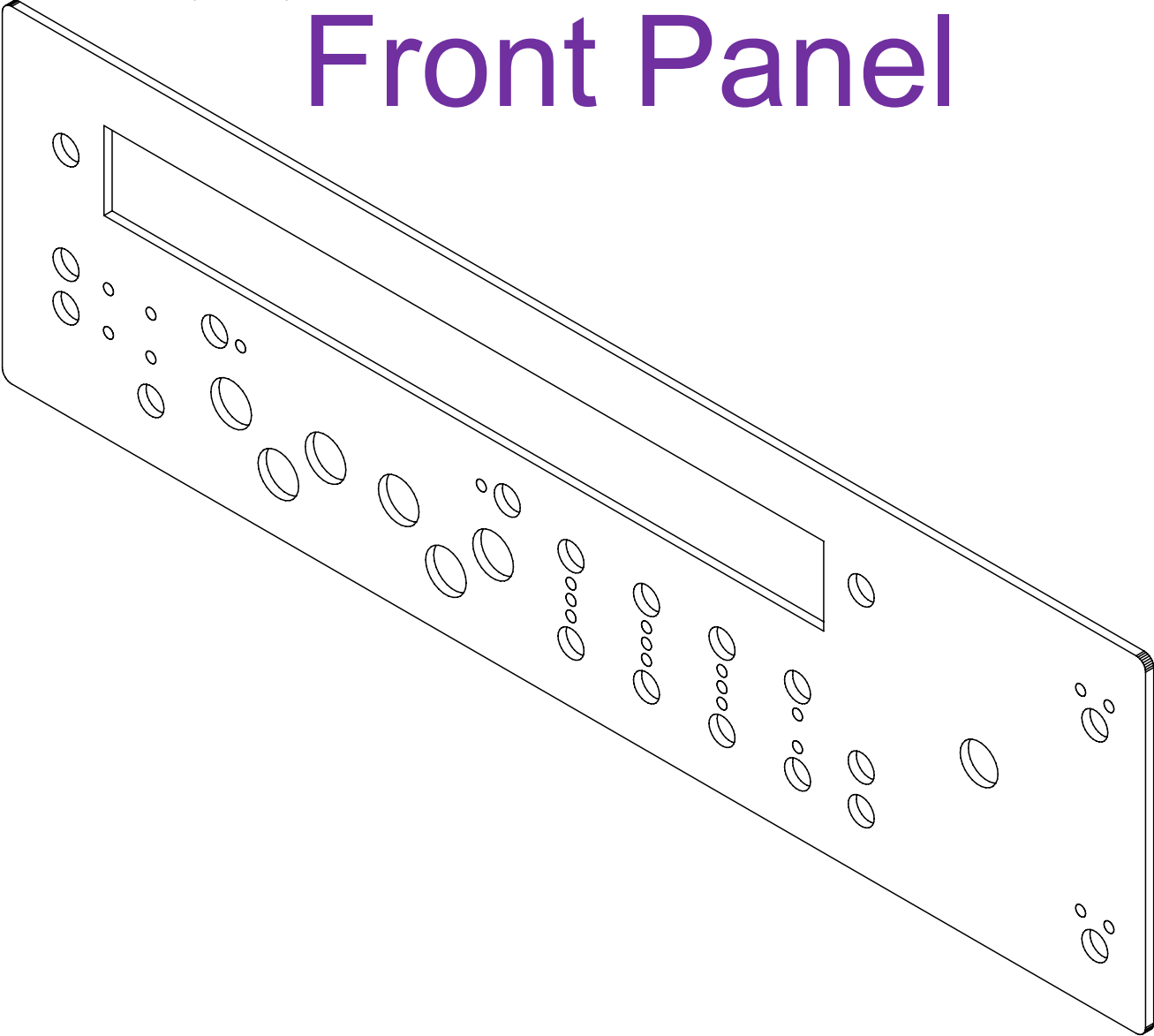
C

D

D

View from Top side (Scale 1:1)

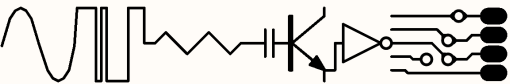
Front Panel

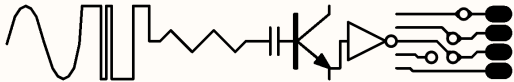


Document Creation Date: 9/6/2025

Design : free_electron

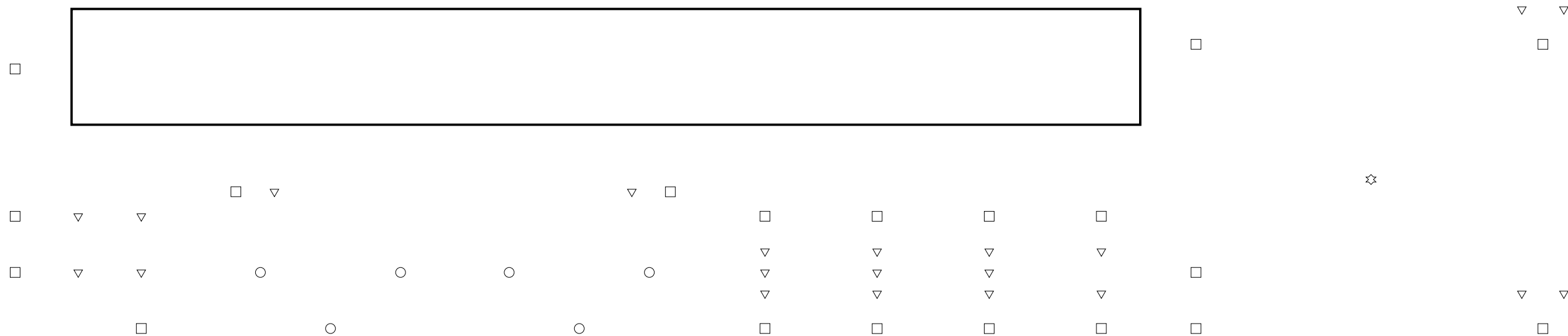
SILICONVALLEYGARAGE



1	2	3	4	5	6
GENERAL					
A	GENERAL				
	1. DO NOT ALTER SUPPLIED COPPER OR DRILL DATA				
	2. NO COPPER BALANCING OR REMOVAL OF UNUSED PADS ALLOWED.				
	3. SILKSCREEN MAY BE CLIPPED / TRIMMED TO EXPOSE COPPER				
	4. PCB DESIGN AND ACCEPTANCE CRITERIA SHALL FOLLOW THE REQUIREMENTS OF IPC-2221, IPC-2222, AND IPC-6012 CLASS 2				
	5. ALL SPECIFICATIONS SHALL BE THE LATEST STANDARDS, UNLESS OTHERWISE NOTED				
B	6. ALL MODIFICATIONS MUST BE COMMUNICATED AND APPROVED IN WRITING.				
	MATERIALS				
	7. MATERIALS SHALL BE ACCORDING TO THE STACKUP DRAWING IN THIS DOCUMENT.				
	8. MATERIAL SHALL HAVE A FLAMABILITY RATING OF UL 94V-0 OR BETTER				
	9. SURFACE FINISH : HASL				
	10. SOLDER MASK COLOR : BLACK				
C	11. SOLDERMASK MAX REGISTRATION ERROR : 0.05mm				
	12. SILKSCREEN COLOR : WHITE				
	STACKUP / IMPEDANCE CONTROL				
	13. THICKNESS LISTED IN LAYER STACK LEGEND REPRESENT FINAL PRESSED VALUES FOR THE PREPREG				
	14. IMPEDANCE CONTROL, IF ANY, SHALL BE PER LISTED TABLE WITH A MAX TOLERANCE OF +/-10%				
	QA, ELECTRICAL TEST AND MARKINGS				
D	15. PCB SHALL BE 100% ELECTRICALLY TESTED FOR SHORTS AND CONTINUITY				
SILICONVALLEYGARAGE				Project CT9042.PrjPcb	
				Version: Variant [No Variations]	
				FABRICATION DRAWING	
1	2	3	4	5	6

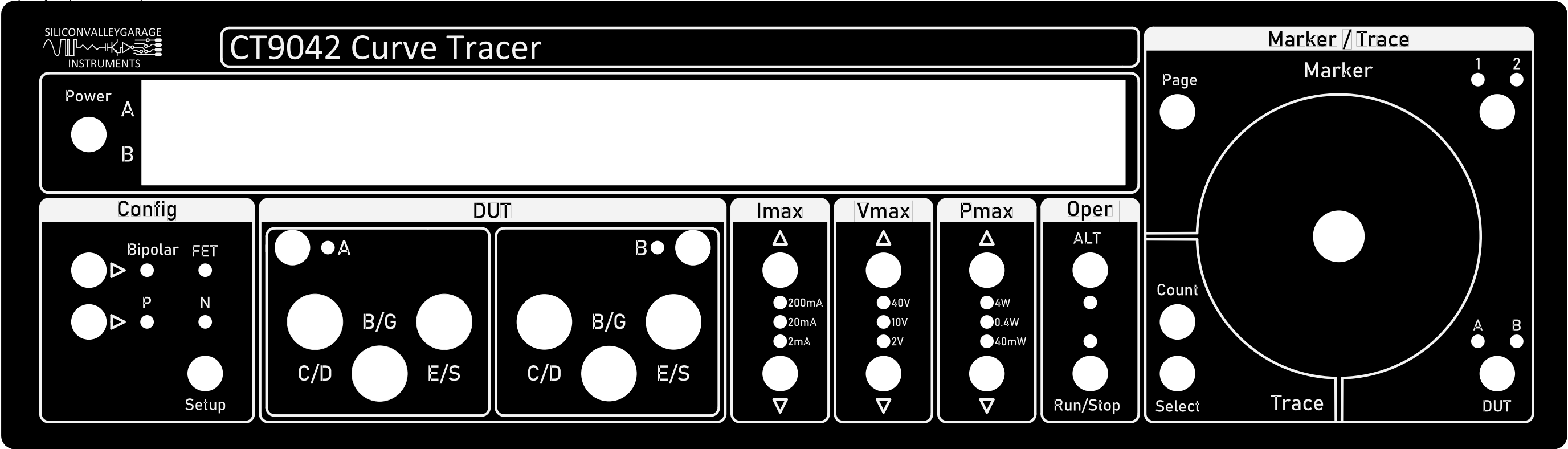
DRILL DRAWING

Drill Drawing View (Scale 1.5:1)



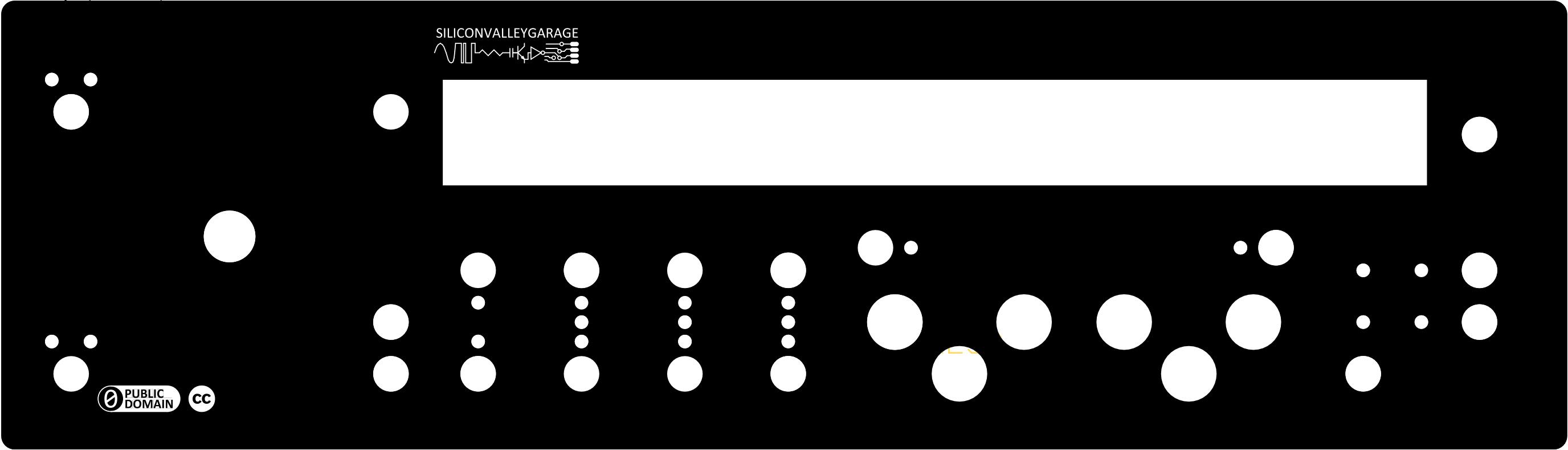
COMPOSITE VIEW FRONT

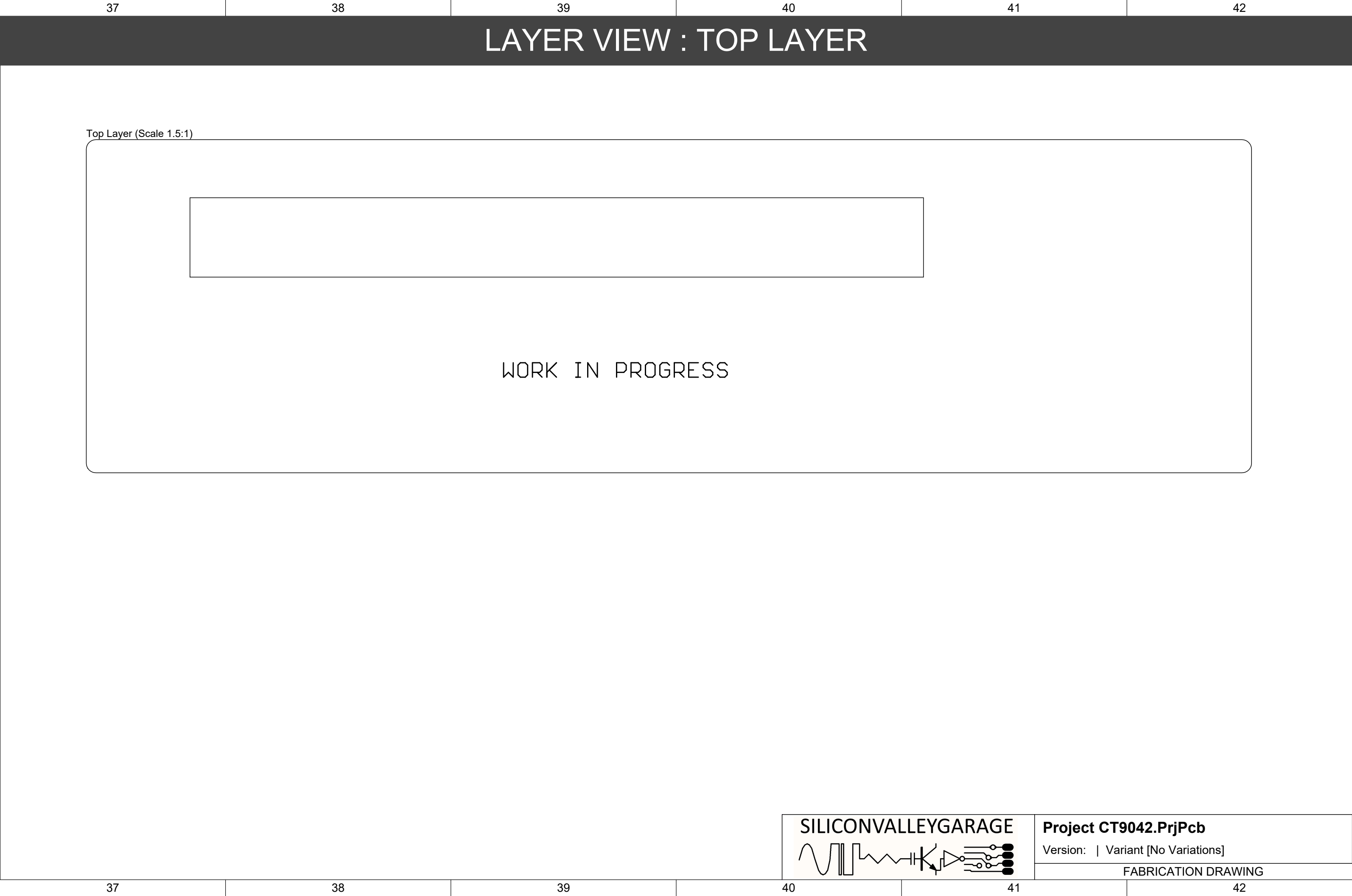
Top Layer (Scale 1.5:1)



COMPOSITE VIEW BACK

Bottom Layer (Scale 1.5:1)





43

44

45

46

47

48

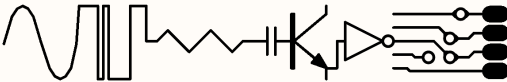
LAYER VIEW : BOTTOM LAYER

Bottom Layer (Scale 1.5:1)



WORK IN PROGRESS

SILICONVALLEYGARAGE



Project CT9042.PrjPcb

Version: | Variant [No Variations]

FABRICATION DRAWING

43

44

45

46

47

48

LAYER VIEW : TOP SOLDER MASK

A

A

Top Solder (Scale 1.5:1)

B

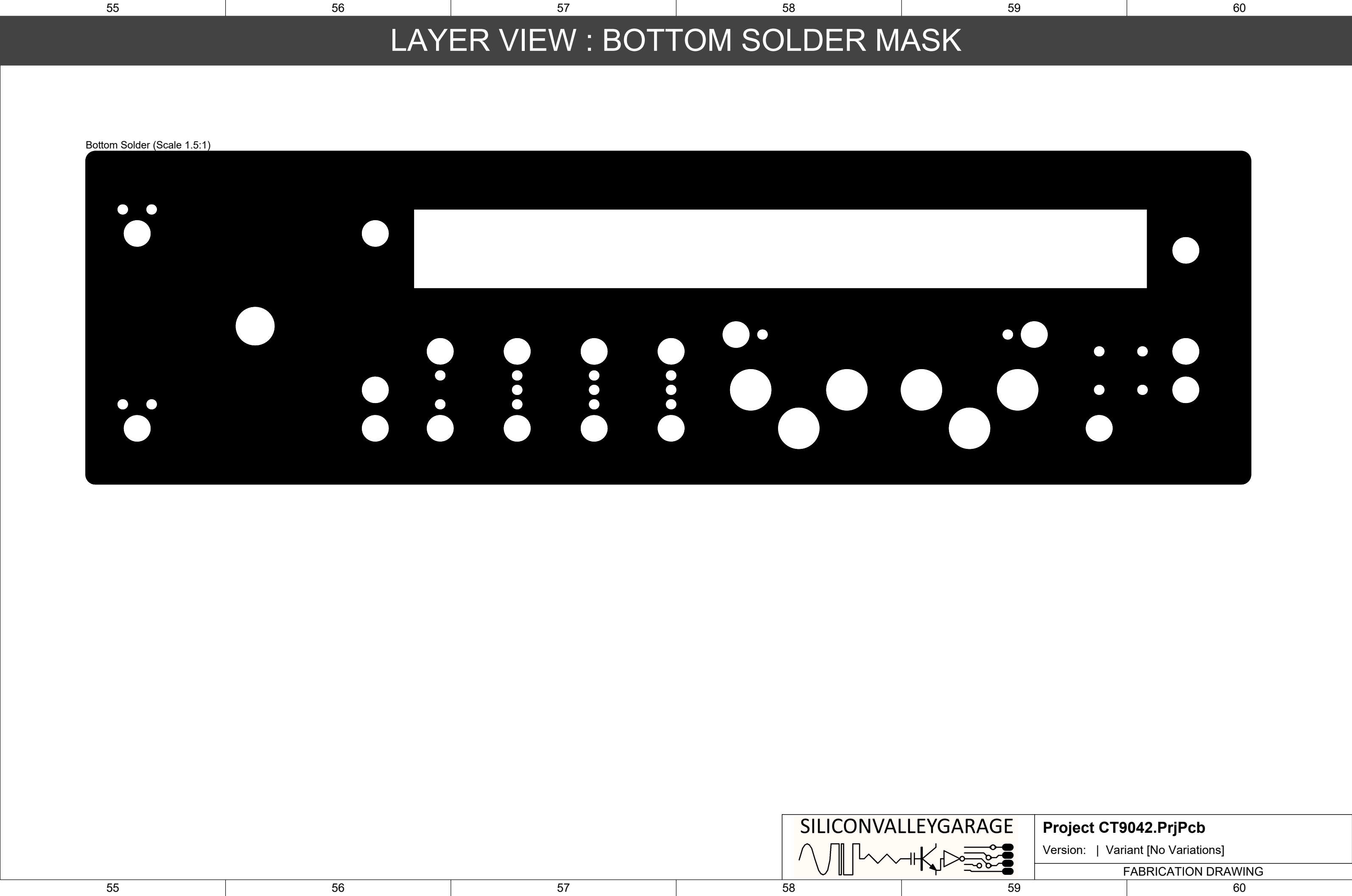
B

C

C

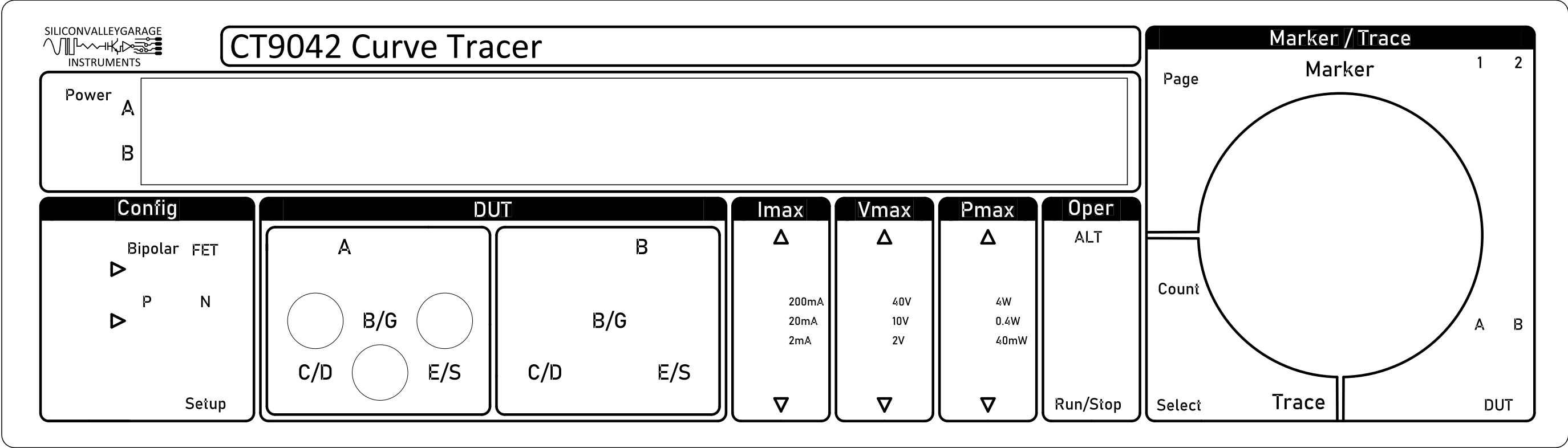
D

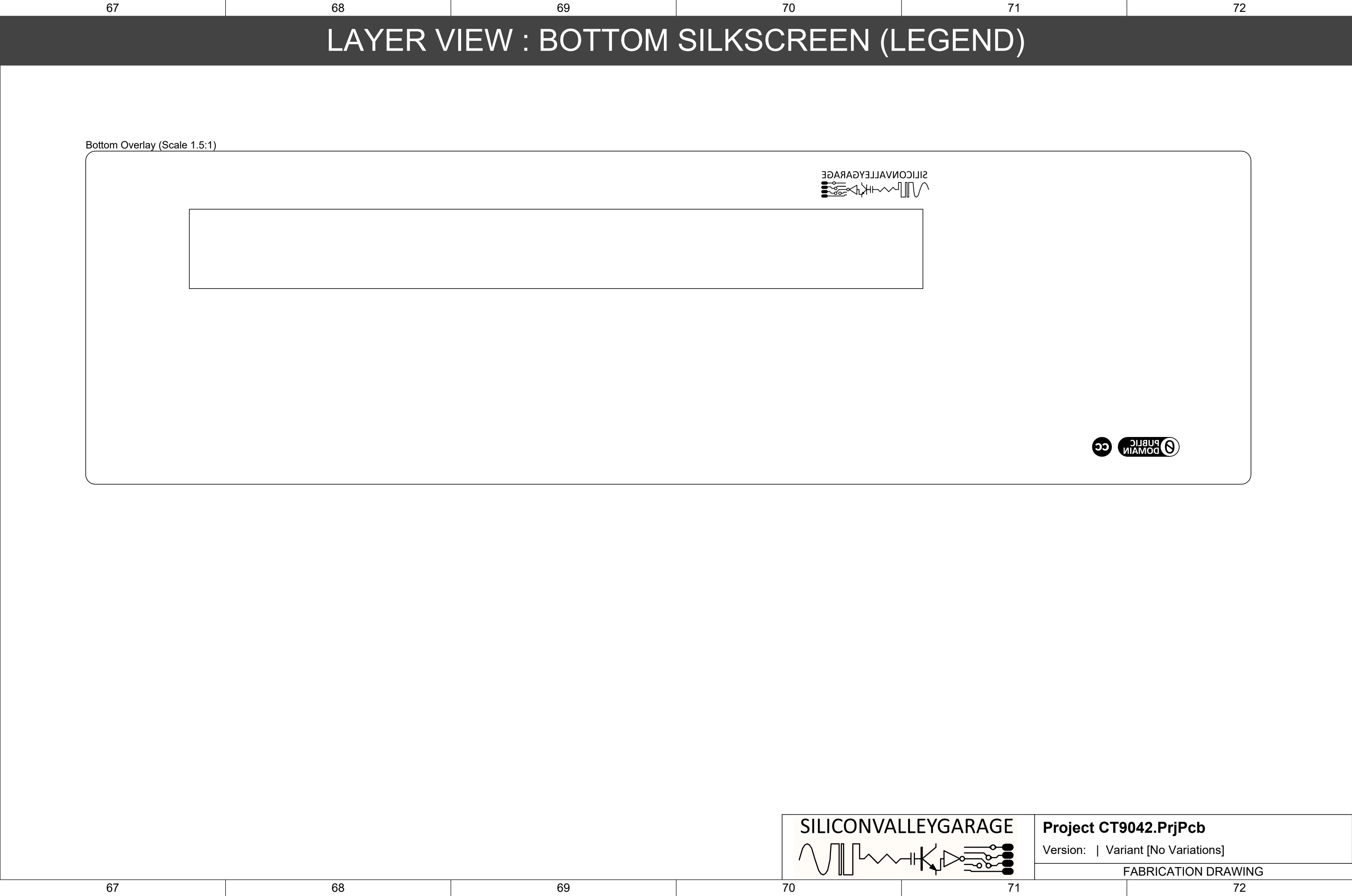
D



LAYER VIEW : TOP SILKSCREEN (LEGEND)

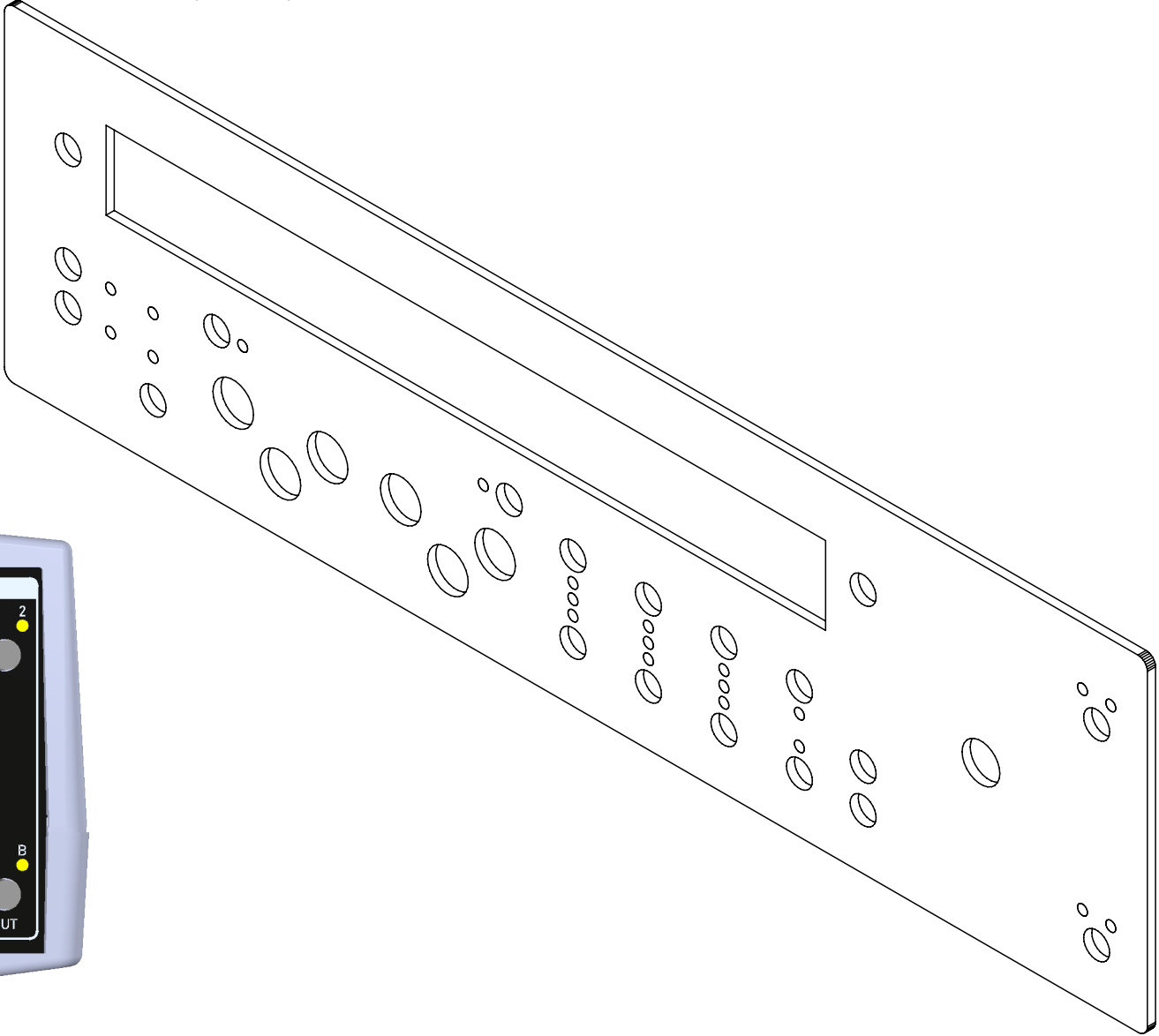
Top Overlay (Scale 1.5:1)





Front Board

View from Top side (Scale 1:1)

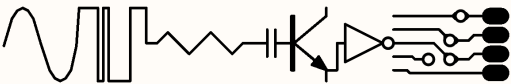


Realistic View



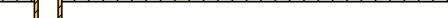
Document Creation Date: 9/6/2025

Design : free_electron

1	2	3	4	5	6
GENERAL					
A	GENERAL				
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	MATERIALS				
	7. MATERIALS SHALL BE ACCORDING TO THE STACKUP DRAWING IN THIS DOCUMENT.				
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	9. SURFACE FINISH : HASL				
	10. SOLDER MASK COLOR : BLACK				
C	11. SOLDERMASK MAX REGISTRATION ERROR : 0.05mm				
	12. SILKSCREEN COLOR : WHITE				
	STACKUP / IMPEDANCE CONTROL				
	13. THICKNESS LISTED IN LAYER STACK LEGEND REPRESENT FINAL PRESSED VALUES FOR THE PREPREG				
	14. IMPEDANCE CONTROL, IF ANY, SHALL BE PER LISTED TABLE WITH A MAX TOLERANCE OF +/-10%				
	QA, ELECTRICAL TEST AND MARKINGS				
D	15. PCB SHALL BE 100% ELECTRICALLY TESTED FOR SHORTS AND CONTINUITY				
SILICONVALLEYGARAGE				Project CT9042.PrjPcb	
				Version: Variant [No Variations]	
				FABRICATION DRAWING	
1	2	3	4	5	6

LAYER STACK

Layer Stack Legend

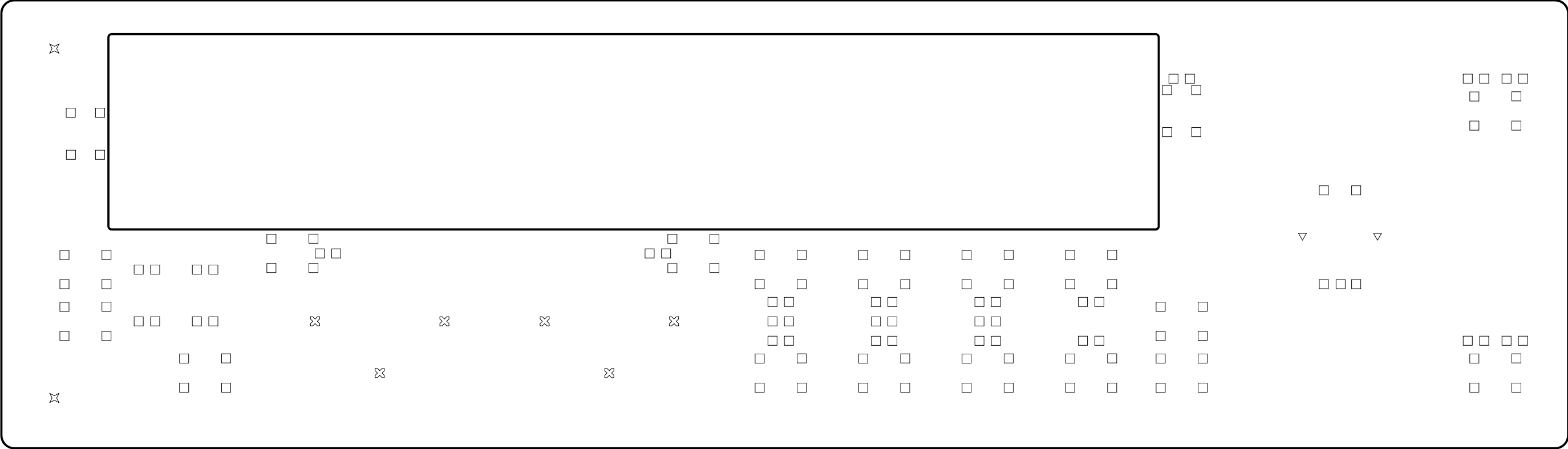
	Material	Layer	Thickness	Dielectric	Material Type	Gerber	Dk	Weight	Constructions	Df	Resin
		Top Overlay			Legend	GTO					
	Surface Material	Top Solder	0.010mm(0.400mil)	Solder Resist	Solder Mask	GTS	3.5				
	Copper	Top Layer	0.036mm(1.400mil)		Signal	GTL		1oz			
	Prepreg		0.102mm(4.000mil)	Core-009	Dielectric		4.5		1-2116	0.02	47%
	CF-004	Layer 1	0.035mm(1.378mil)		Signal	G1		1oz			
			1.450mm(57.087mil)	FR-4	Dielectric		4.8				
	CF-004	Layer 2	0.035mm(1.378mil)		Signal	G2		1oz			
	Prepreg		0.102mm(4.000mil)	Core-009	Dielectric		4.5		1-2116	0.02	47%
	Copper	Bottom Layer	0.036mm(1.400mil)		Signal	GBL		1oz			
	Surface Material	Bottom Solder	0.010mm(0.400mil)	Solder Resist	Solder Mask	GBS	3.5				
		Bottom Overlay			Legend	GBO					
Total thickness: 1.815mm(71.443mil)											

Drill Table

Symbol	Count	Hole Size	Plated	Hole Type	Drill Layer Pair	Via / Pad	Pad Shape	Description	Hole Tolerance	Via Type	Via Feature
□	125	1.000mm(39.370mil)	Plated	Round	Top Layer - Bottom Layer	Pad	Rounded				
▽	2	2.600mm(102.362mil)	Plated	Round	Top Layer - Bottom Layer	Pad	Rounded				
✕	2	3.200mm(125.984mil)	Plated	Round	Top Layer - Bottom Layer	Pad	Rounded				
⊗	6	10.000mm(393.701mil)	Non-Plated	Round	Top Layer - Bottom Layer	Pad	Rounded				
	135 Total										

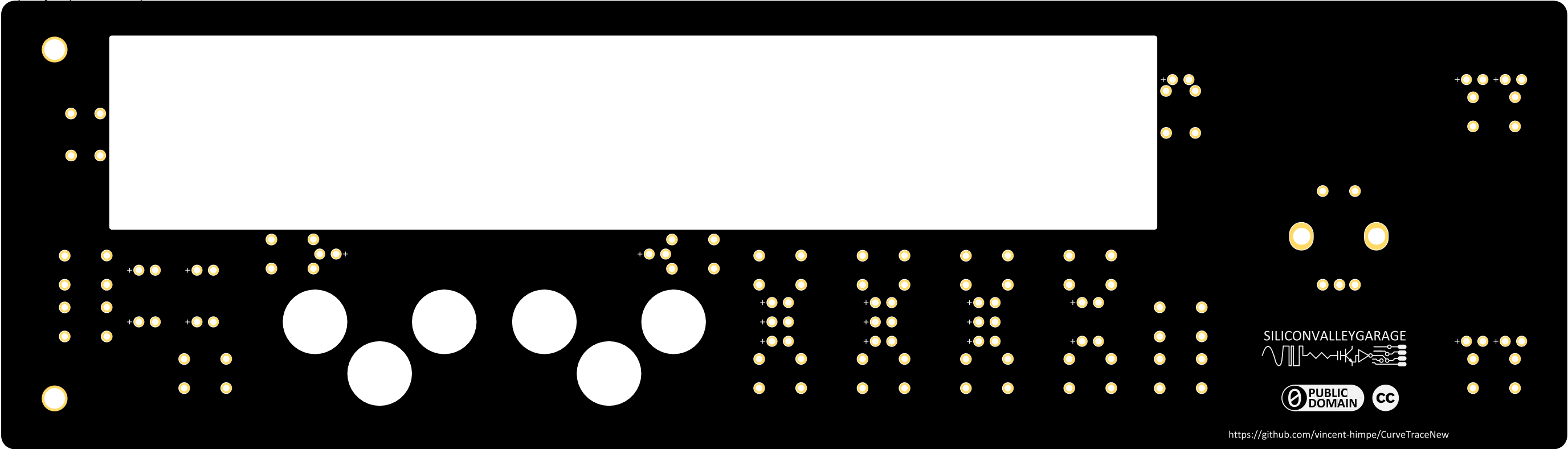
DRILL DRAWING

Drill Drawing View (Scale 1.5:1)



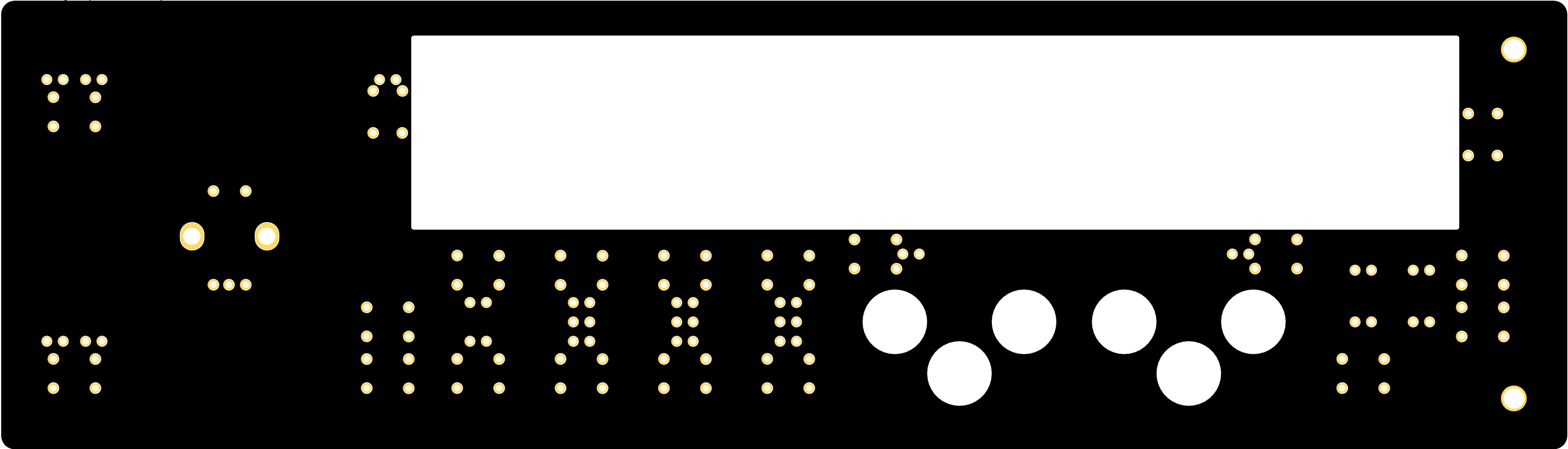
COMPOSITE VIEW FRONT

Top Layer (Scale 1.5:1)



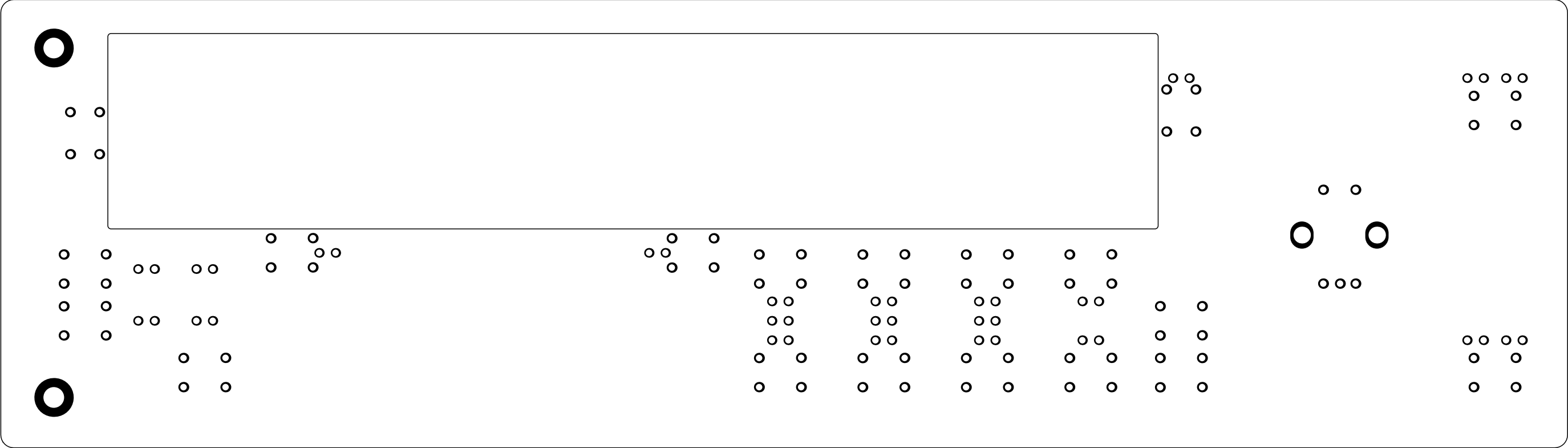
COMPOSITE VIEW BACK

Bottom Layer (Scale 1.5:1)



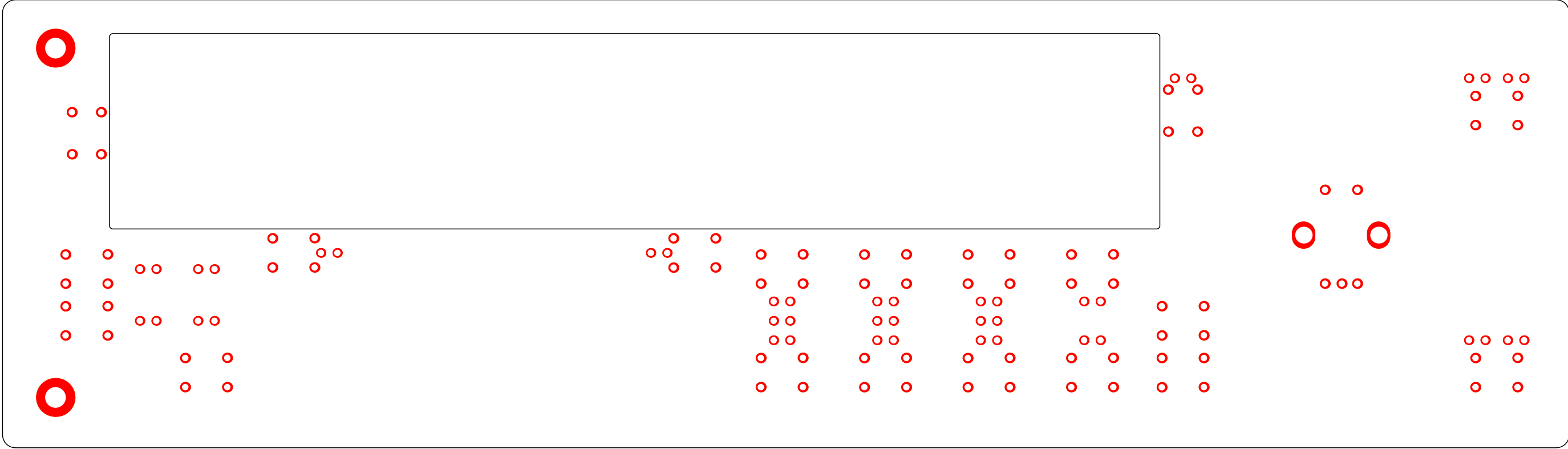
LAYER VIEW : TOP LAYER

Top Layer (Scale 1.5:1)



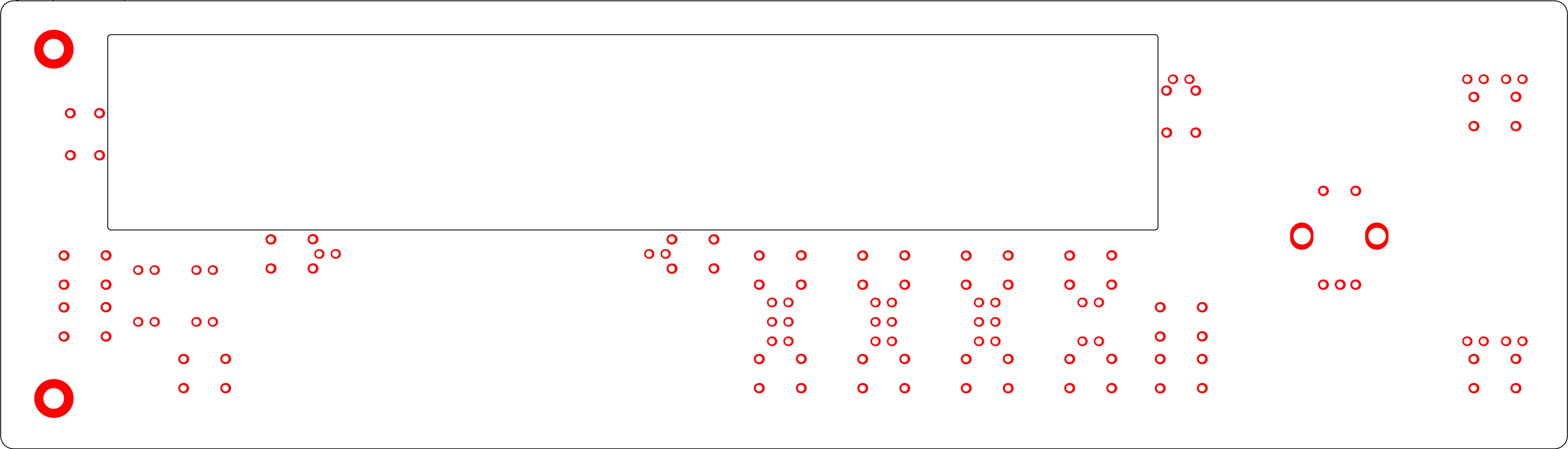
LAYER VIEW : MID LAYER 1

Layer 1 (Scale 1.5:1)



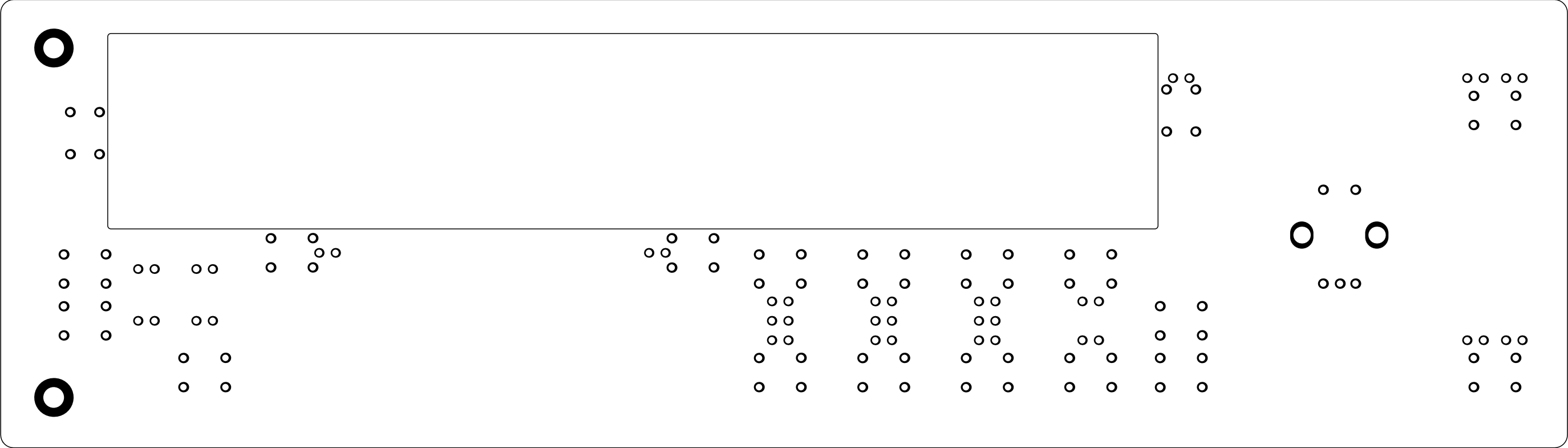
LAYER VIEW : MID LAYER 2

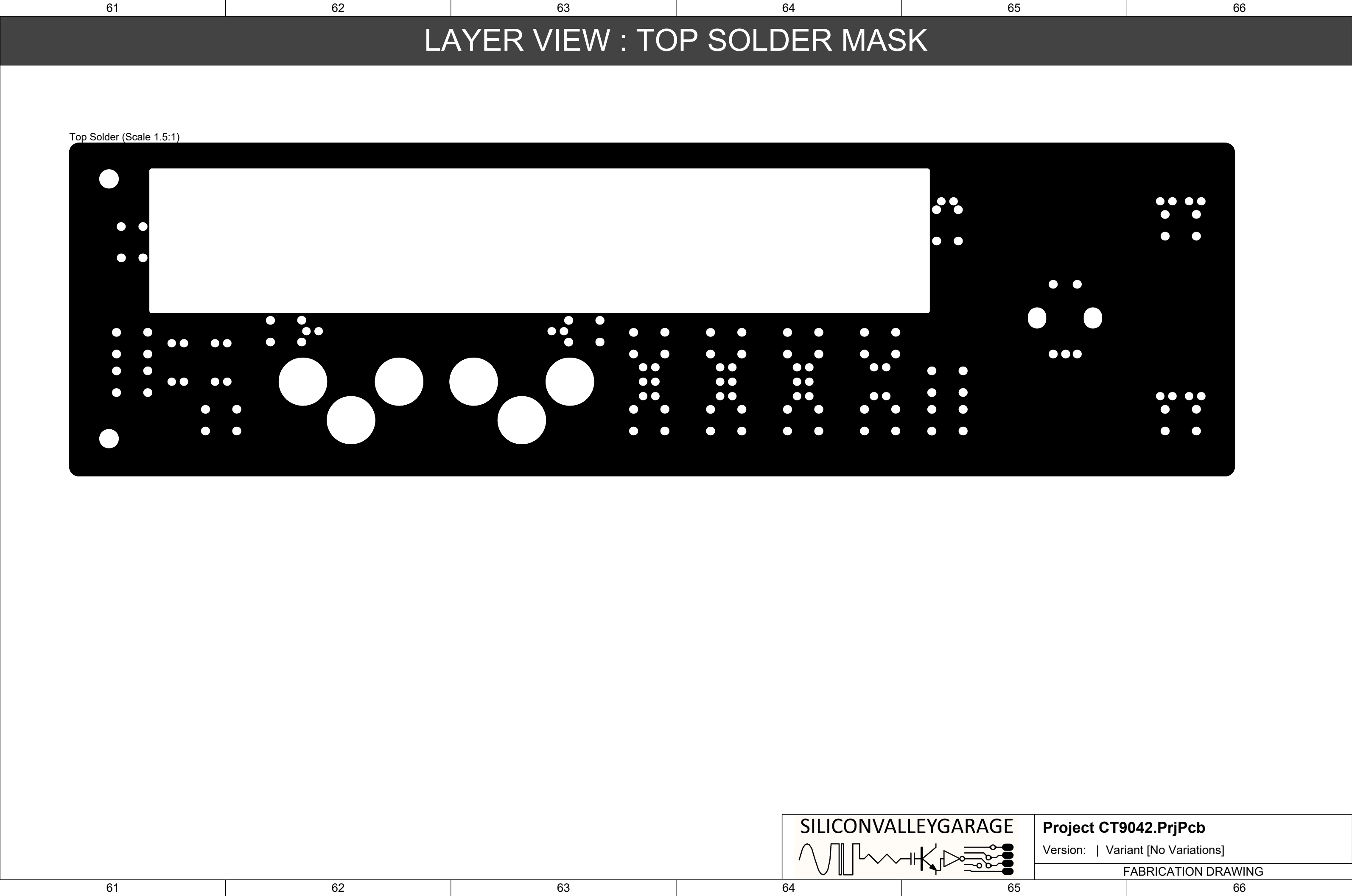
Layer 2 (Scale 1.5:1)



LAYER VIEW : BOTTOM LAYER

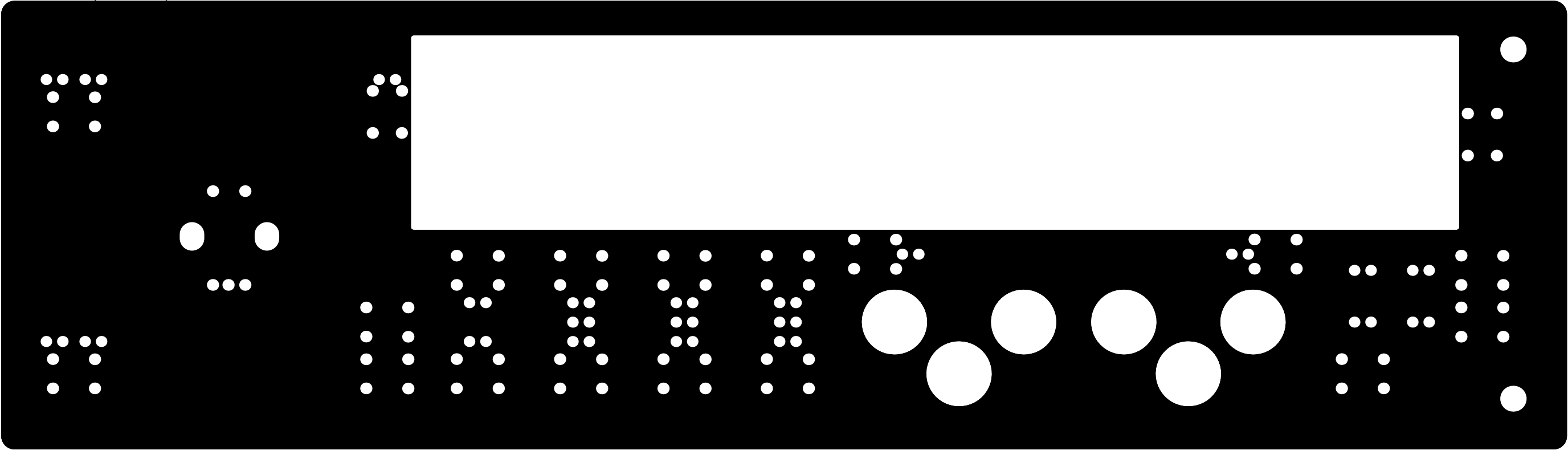
Bottom Layer (Scale 1.5:1)





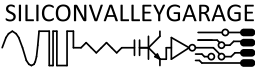
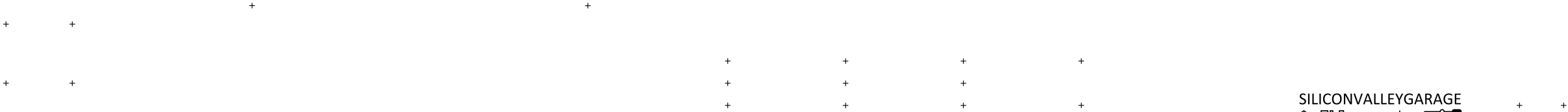
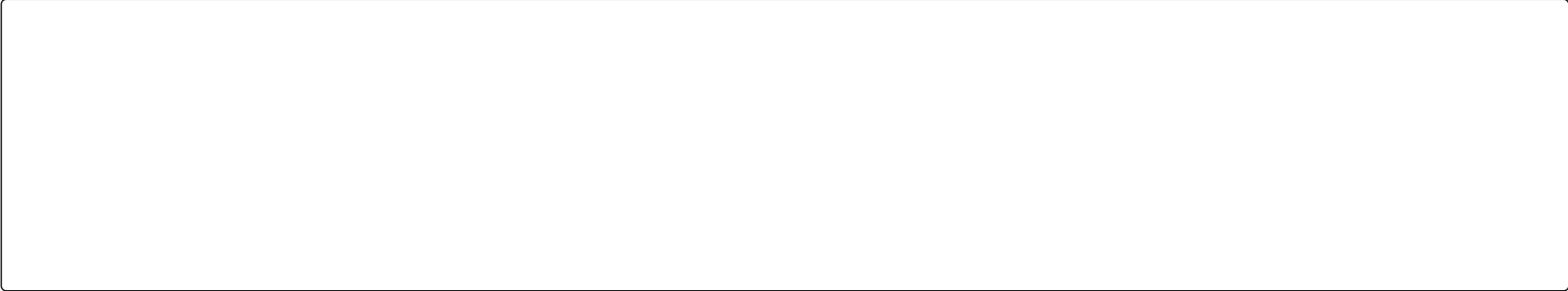
LAYER VIEW : BOTTOM SOLDER MASK

Bottom Solder (Scale 1.5:1)

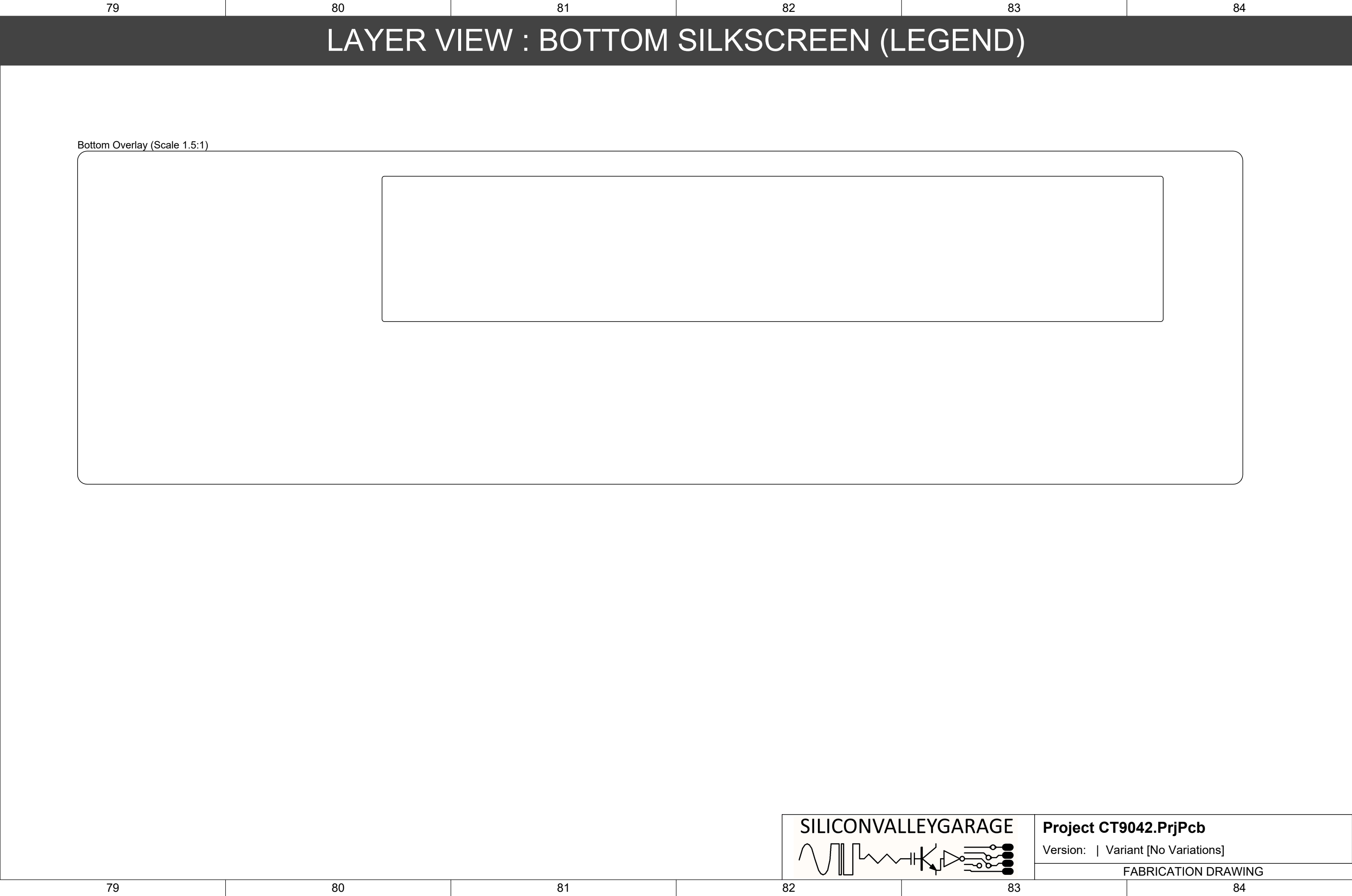


LAYER VIEW : TOP SILKSCREEN (LEGEND)

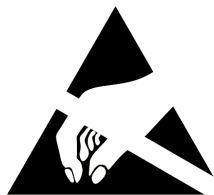
Top Overlay (Scale 1.5:1)



<https://github.com/vincent-himpe/CurveTraceNew>



GENERAL



Unless otherwise specified the following rules apply:

- 1. DO NOT DEVIATE FROM ARTWORK OR BOM WITHOUT PRIOR AUTHORIZATION.
- 2. ASSEMBLE AND INSPECT PER IPC-610 CLASS 2

Bill of Materials and Material Handling

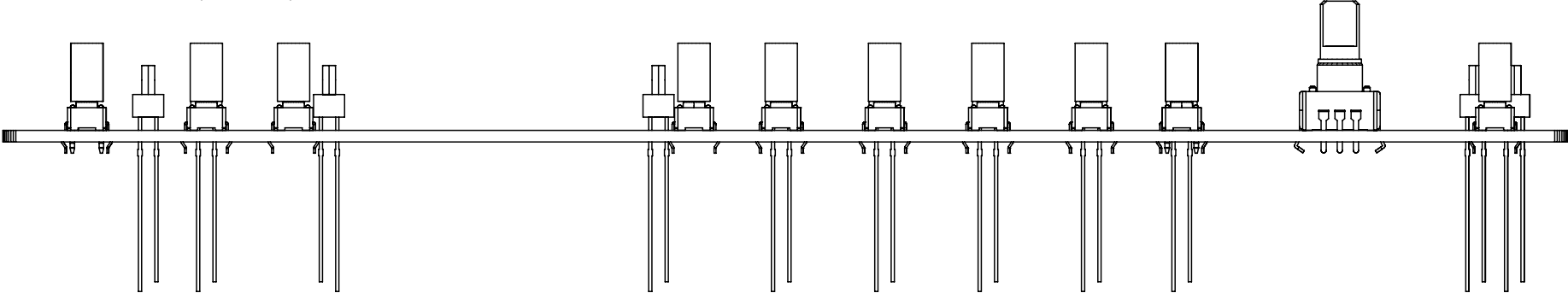
- 3. THE BOM CONTAINED IN THIS DOCUMENT IS AS-BUILT. NON-INSTALLED PARTS HAVE BEEN REMOVED. ADDITIONAL BOM FORMATS ARE AVAILABLE IN THE PROJECT FILES
- 4. ANY PART SUBSTITUTIONS MUST BE APPROVED IN WRITING BEFORE ASSEMBLY
- 5. ALL MATERIALS MUST BE PROCURED FROM MANUFACTURER AUTHORIZED DISTRIBUTORS OR THE ORIGINAL MANUFACTURER
- 6. ALL COMPONENTS AND BOARDS TO BE HANDLED AND STORED ACCORDING TO IPC GUIDELINES
- 7. ESD CONTROL PER IPC RULES

Soldering

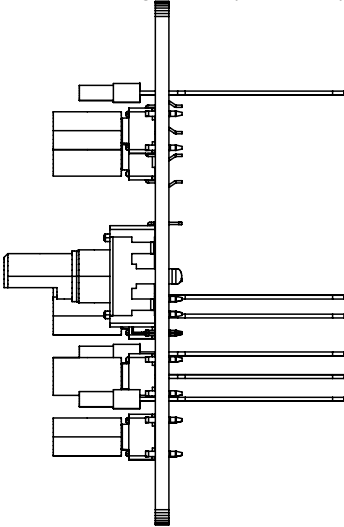
- 8. SOLDERING TO BE DONE USING SN37PB63 ALLOY USING ALLOY MANUFACTURER RECOMMENDED NO-CLEAN FLUX
- 9. BGA COMPONENTS WITH LEAD-FREE CONNECTIONS NEED TO BE REBALLED WITH SN63PB37. MIXING OF ALLOYS IS NOT PERMITTED.
- 10. SOLDERING PREFERRABLY TO BE DONE USING NITROGEN ATMOSPHERE
- 11. SURPLUS COMPONENTS TO VACUUM SEALED WITH DESSICANT IN ANTISTATIC BAGS
- 12. INCOMING MATERIAL (BOARDS AND COMPONENTS) NEEDS TO BE INSPECTED FOR HUMIDITY AND BAKED IF NEEDED PRIOR TO USE.
- 13. MANUAL REWORK / TOUCHUP TO BE DONE USING SAME ALLOY AND APPROPRIATE FLUX. FLUX MUST BE REMOVED.

2D VIEW

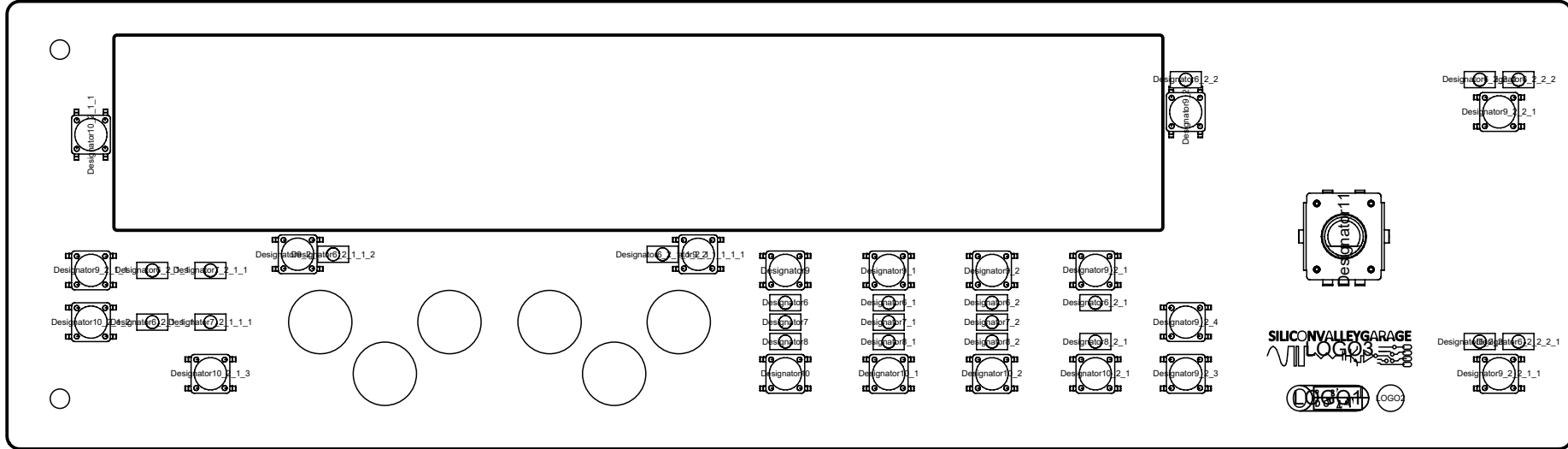
View from Front side (Scale 1:1)



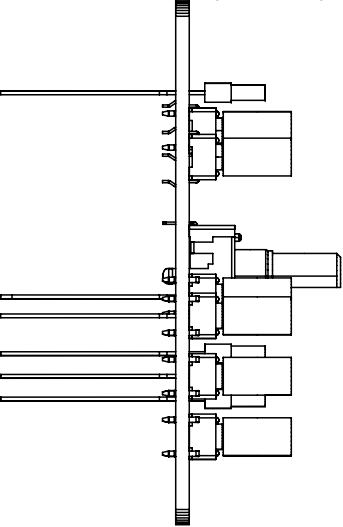
View from Right side (Scale 1:1)



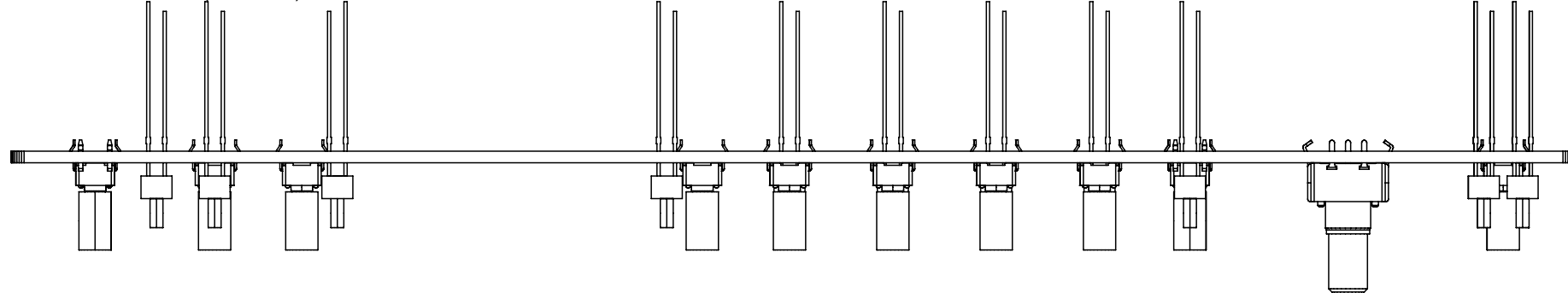
View from Top side (Scale 1:1)



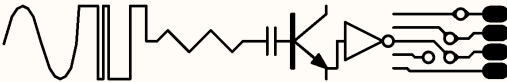
View from Left side (Scale 1:1)



View from Back side (Scale 1:1)



SILICONVALLEYGARAGE



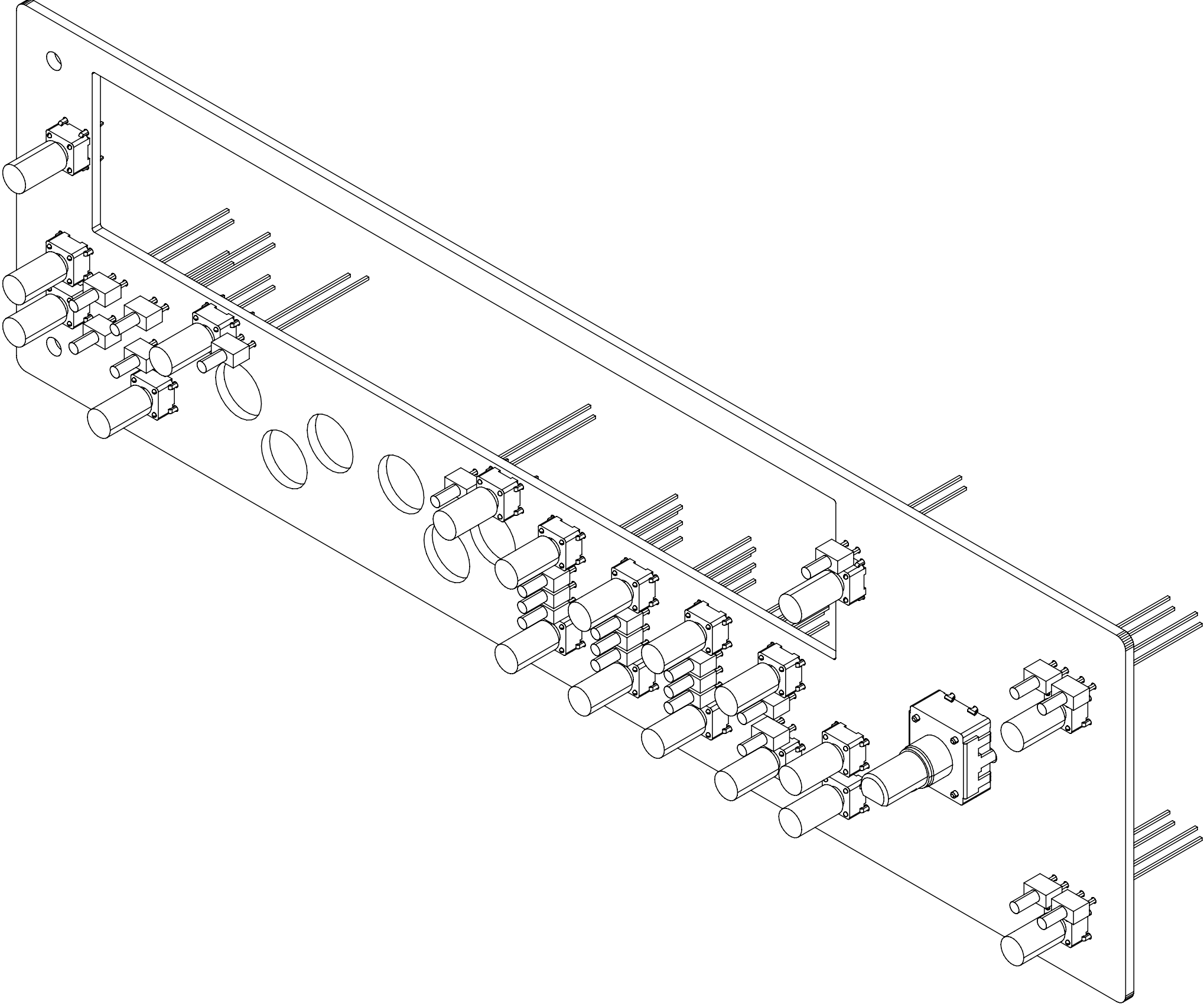
Project CT9042.PrjPcb

Version: | Variant [No Variations]

ASSEMBLY DRAWING

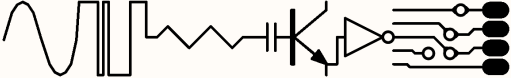
3D VIEW

View from Top side (Scale 1.45:1)



Bill Of Materials

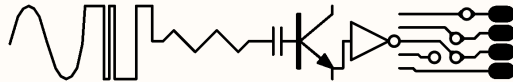
Quantity	Designator	Description	LCSC	MOUSER
40	C1, C4, C6, C8, C10, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C22, C24, C25, C29, C30, C31, C32, C36, C37, C68, C69, C70, C71, C73, C74, C75, C76, C77, C78, C79, C80, C83, C84, C?	CAPACITOR,CERAMIC,100nF,50V,X7R,0603	C127833	80-C0603C104K5R
5	C2, C3, C5, C7, C9	CAPACITOR,CERAMIC,10uF,25V,X5R,0805	C3039694	187-CL21B106KAYQNNE
2	C21, C33	CAPACITOR,CERAMIC,10nF,50V,X7R,0603		
3	C23, C34, C217	CAPACITOR,CERAMIC,100pF,50V,C0G,0603		
2	C26, C28	CAPACITOR,ALU,47uF,100V,TH,P5mm,D10mm,H14mm		
2	C27, C234	CAPACITOR,CERAMIC,10pF,50V,C0G,0603		80-C0603C100J5GAC
2	C35, C47	CAPACITOR,CERAMIC,1nF,50V,C0G,0603		
6	C38, C44, C48, C52, C56, C60	CAPACITOR,ALU,47uF,35V,TH,P0.25mm,D6.3mm,H12.5mm		
1	C39	CAPACITOR,CERAMIC,100nF,100V,X7R,0805		
1	C40	CAPACITOR,ALU,2200uF,100V,TH,P10mm,D35mm,H32mm,SNAP		
4	C41, C43, C45, C46	CAPACITOR,CERAMIC,10nF,100V,C0G,0805		
1	C42	CAPACITOR,CERAMIC,4u7,10V,X5R,0603	C1705	187-CL10A475KP8NNNC
1	C72	CAPACITOR,POLYPROP,100nF,100V		
1	C81	CAPACITOR,CERAMIC,1n5,50V,X7R,0603		
21	D1, D2, D3, D4, D5, D6, D7, D8, D9, D10, D11, D12, D13, D14, D15, D16, D17, D18, D19, D20, D21	LED,TH,GREEN,42mCd,3MM		78-VLMPG33N1P2
7	D22, D23, D28, D29, D30, D38, D87	DIODE,75V,250mA,SOT23,500pA ULTRA LOW LEAKAGE		
8	D24, D25, D26, D27, D39, D43, D44, D45	DIODE,RECTIFIER,75V,0.3A,MMBD4148,SOT23		
17	D31, D32, D33, D36, D37, D41, D42, D46, D80, D81, D82, D83, D84, D85, D86, D?	DIODE,RECTIFIER,75V,0.3A,BAS16,SOT23		
10	D34, D35, D60, D63, D67, D68, D70, D73, D215, D231	DIODE,RECTIFIER,1000V,1A,ONSEMI,NRVA4007,SMA		
1	DISP1	DISPLAY,LCD,ALPHA,2X40CHAR,HD44780,YELLOW BL		
1	HS9	HEATSINKSTAMPED,CLIP-ON,BOARDMOUNT,3W,25C/W,CUI,HSS01-B20-CP		
2	J1, J2			
1	J3	CONN,2X3,MOLEX MICROFIT,TH,VERT		
1	J4	CONN,1X2,MOLEX, MICROFIT,VERT		
1	J9	CONN,POWER,IEC,SWITCH,FUSE ASSEMBLY,PANELMOUNT		
1	JX			
1	JY			
1	JZ			
7	K1, K2, K3, K4, K5, K7, K10	RELAY,DPDT,1A,250VAC,220VDC,5V COIL,KEMET,UA2-5NJ		
2	K6, K9	RELAY,OPTOMOS,6A,60V,AC,IXYS,CPC1907B,SMD		
2	K8, K11	RELAY,OPTOMOS,1.5A,100V,AC,IXYS,LCA701S,SMD		
1	K12	RELAY,DPDT,5V COIL,250VAC,3A		
13	Q1, Q2, Q4, Q6, Q7, Q8, Q9, Q10, Q11, Q12, Q13, Q14, Q24	XSTR,NPN,45V,0.1A,,BC847B,SOT23		
5	Q3, Q5, Q18, Q19, Q201	XSTR,PNP,SOT23,BC857,SOT23		
5	Q15, Q20, Q21, Q22, Q23	XSTR,NMOS,30V,4.3A,SI2336DS-T1-BE3.SOT23		
2	Q16, Q25	XSTR,PMOS,100V,5.6A,DMP10H400SK3-13,DPAK		
1	Q17	XSTR,NMOS,100V,12A,DMN10H170SK3-13,DPAK		
50	R1, R2, R3, R4, R29, R30, R39, R42, R51, R58, R60, R61, R67, R68, R70, R71, R76, R101, R102, R110, R120, R124, R125, R127, R131, R132, R133, R138, R139, R140, R141, R142, R144, R145, R146, R147, R148, R151, R152, R153, R154, R155, R156, R242, R243, R277, R279, R280, R?	RESISTOR,10K,0.1%,100mW,0603 (1608)		
1	R5	RES,TRIM,SMD,500R		652-TC33X-2-101E
1	R6	RESISTOR,GENERIC,0805		
27	R7, R8, R9, R10, R11, R12, R13, R14, R15, R16, R17, R18, R19, R20, R21, R22, R23, R24, R25, R26, R27, R28, R41, R119, R122, R126, R128	RESISTOR,511R,0.1%,100mW,0603 (1608)		

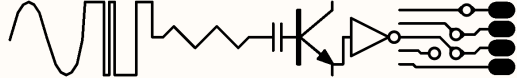
SILICONVALLEYGARAGE

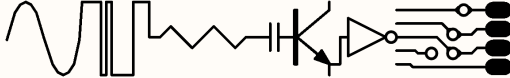
Project CT9042.PrjPcb

Version: | Variant [No Variations]

ASSEMBLY DRAWING

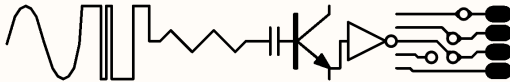
DESIGNATORS FRONT					
A					
B					
C					
D					
				SILICONVALLEYGARAGE 	
				Project CT9042.PrjPcb Version: Variant [No Variations]	
				ASSEMBLY DRAWING	

1	2	3	4	5	6
DESIGNATORS BACK					
A					
B					
C					
D					
				SILICONVALLEYGARAGE Project CT9042.PrjPcb Version: Variant [No Variations] ASSEMBLY DRAWING	
1	2	3	4	5	6

PASTE MASK TOP					
A					
B					
C					
D					
				SILICONVALLEYGARAGE Project CT9042.PrjPcb Version: 1.0 Variant [No Variations] ASSEMBLY DRAWING	

1	2	3	4	5	6
PASTE MASK BACK					
A					A
B					B
C					C
D					D
1	2	3	4	5	6

SILICONVALLEYGARAGE



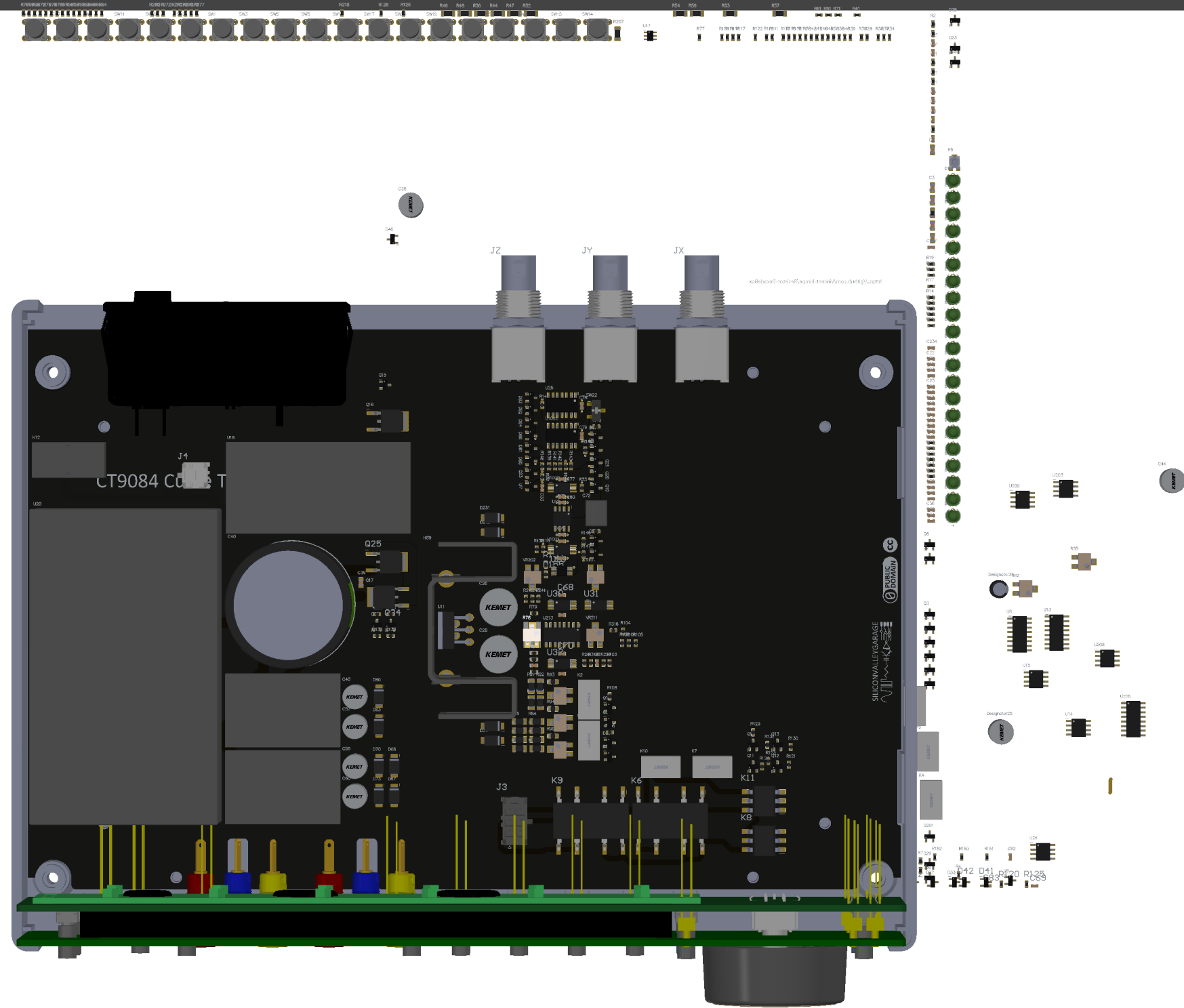
Project CT9042.PrjPcb

Version: | Variant [No Variations]

ASSEMBLY DRAWING

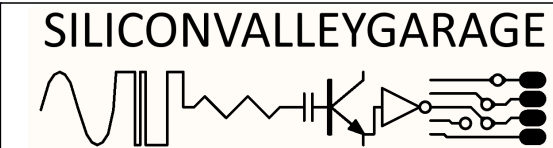
CT9042.PrjPcb

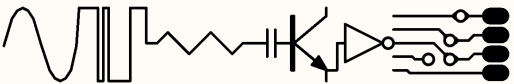
Realistic View



Document Creation Date: 9/6/2025

Design : free_electron



1	2	3	4	5	6
GENERAL					
A	GENERAL				
	1. DO NOT ALTER SUPPLIED COPPER OR DRILL DATA				
	2. NO COPPER BALANCING OR REMOVAL OF UNUSED PADS ALLOWED.				
	3. SILKSCREEN MAY BE CLIPPED / TRIMMED TO EXPOSE COPPER				
	4. PCB DESIGN AND ACCEPTANCE CRITERIA SHALL FOLLOW THE REQUIREMENTS OF IPC-2221, IPC-2222, AND IPC-6012 CLASS 2				
	5. ALL SPECIFICATIONS SHALL BE THE LATEST STANDARDS, UNLESS OTHERWISE NOTED				
B	6. ALL MODIFICATIONS MUST BE COMMUNICATED AND APPROVED IN WRITING.				
	MATERIALS				
	7. MATERIALS SHALL BE ACCORDING TO THE STACKUP DRAWING IN THIS DOCUMENT.				
	8. MATERIAL SHALL HAVE A FLAMABILITY RATING OF UL 94V-0 OR BETTER				
	9. SURFACE FINISH : HASL				
	10. SOLDER MASK COLOR : BLACK				
C	11. SOLDERMASK MAX REGISTRATION ERROR : 0.05mm				
	12. SILKSCREEN COLOR : WHITE				
	STACKUP / IMPEDANCE CONTROL				
	13. THICKNESS LISTED IN LAYER STACK LEGEND REPRESENT FINAL PRESSED VALUES FOR THE PREPREG				
	14. IMPEDANCE CONTROL, IF ANY, SHALL BE PER LISTED TABLE WITH A MAX TOLERANCE OF +/-10%				
	QA, ELECTRICAL TEST AND MARKINGS				
D	15. PCB SHALL BE 100% ELECTRICALLY TESTED FOR SHORTS AND CONTINUITY				
SILICONVALLEYGARAGE				Project CT9042.PrjPcb	
				Version: Variant [No Variations]	
				FABRICATION DRAWING	
1	2	3	4	5	6

LAYER STACK

Layer Stack Legend

Material	Layer	Thickness	Dielectric	Material Type	Gerber	Dk	Weight	Constructions	Df	Resin
	Top Overlay			Legend	GTO					
Surface Material	Top Solder	0.010mm(0.400mil)	Solder Resist	Solder Mask	GTS	3.5				
Copper	Top Layer	0.036mm(1.400mil)		Signal	GTL		1oz			
Prepreg		0.102mm(4.000mil)	Core-009	Dielectric		4.5		1-2116	0.02	47%
CF-004	Layer 1	0.035mm(1.378mil)		Signal	G1		1oz			
		1.450mm(57.087mil)	FR-4	Dielectric		4.8				
CF-004	Layer 2	0.035mm(1.378mil)		Signal	G2		1oz			
Prepreg		0.102mm(4.000mil)	Core-009	Dielectric		4.5		1-2116	0.02	47%
Copper	Bottom Layer	0.036mm(1.400mil)		Signal	GBL		1oz			
Surface Material	Bottom Solder	0.010mm(0.400mil)	Solder Resist	Solder Mask	GBS	3.5				
	Bottom Overlay			Legend	GBO					
Total thickness: 1.815mm(71.443mil)										

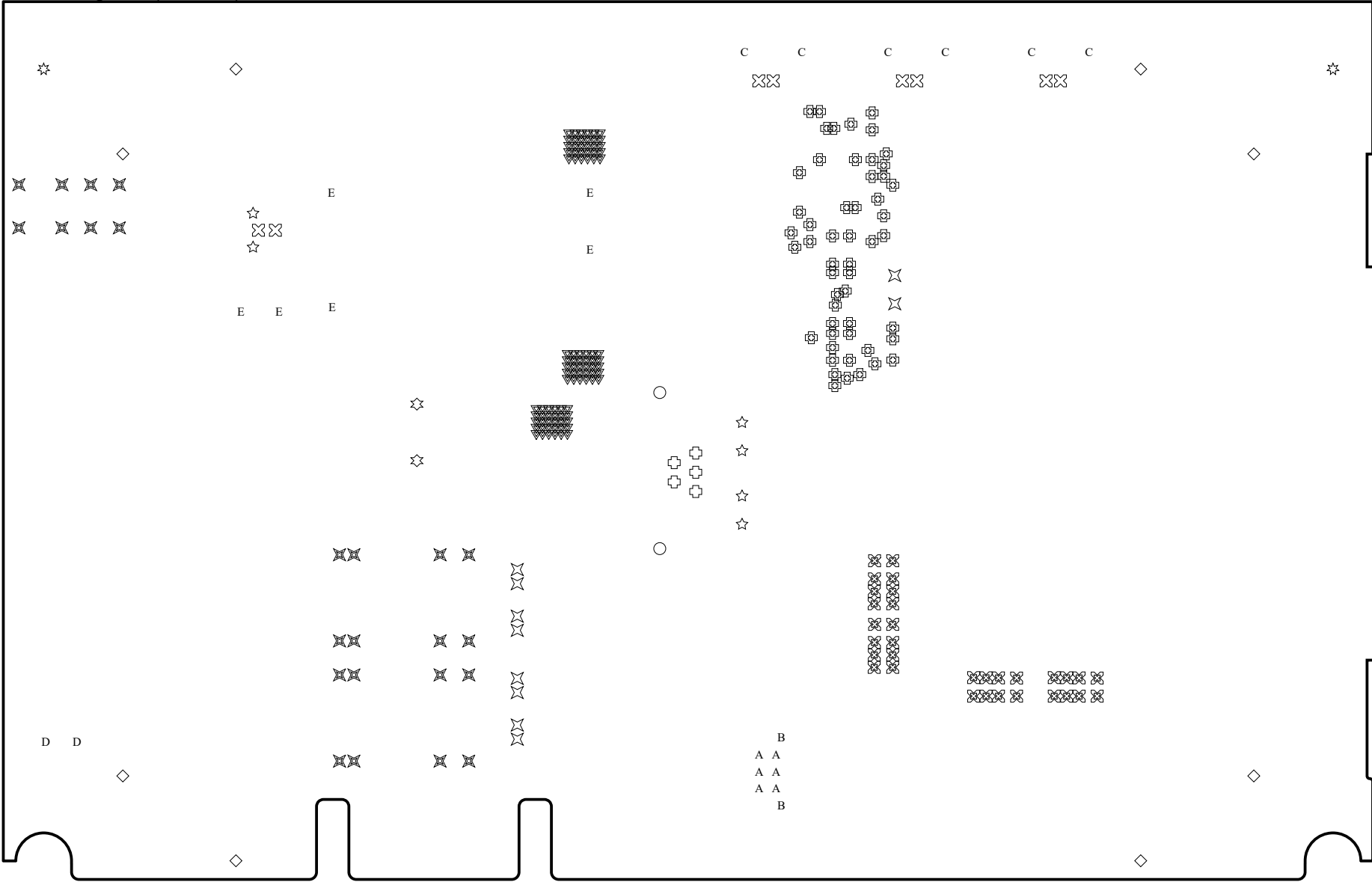
DRILL LEGEND

Drill Table

Symbol	Count	Hole Size	Plated	Hole Type	Drill Layer Pair	Via / Pad	Pad Shape	Description	Hole Tolerance	Via Type	Via Feature
	53	0.200mm(7.874mil)	Plated	Round	Top Layer - Bottom Layer	Via					
	90	0.254mm(10.000mil)	Plated	Round	Top Layer - Bottom Layer	Via					
	56	0.700mm(27.559mil)	Non-Plated	Round	Top Layer - Bottom Layer	Pad	Rounded				
	18	0.800mm(31.496mil)	Non-Plated	Round	Top Layer - Bottom Layer	Pad	Rounded				
	6	1.000mm(39.370mil)	Non-Plated	Round	Top Layer - Bottom Layer	Pad	Rounded				
	126	1.000mm(39.370mil)	Plated	Round	Top Layer - Bottom Layer	Pad	(Mixed)				
B	2	1.020mm(40.158mil)	Non-Plated	Round	Top Layer - Bottom Layer	Pad	Rounded				
A	6	1.020mm(40.158mil)	Plated	Round	Top Layer - Bottom Layer	Pad	(Mixed)				
	24	1.100mm(43.307mil)	Non-Plated	Round	Top Layer - Bottom Layer	Pad	Rounded				
	5	1.200mm(47.244mil)	Plated	Round	Top Layer - Bottom Layer	Pad	Rounded				
E	6	1.300mm(51.181mil)	Non-Plated	Round	Top Layer - Bottom Layer	Pad	Rounded				
C	6	2.000mm(78.740mil)	Plated	Round	Top Layer - Bottom Layer	Pad	Rounded				
	2	2.100mm(82.677mil)	Non-Plated	Round	Top Layer - Bottom Layer	Pad	Rounded				
D	2	2.400mm(94.488mil)	Non-Plated	Round	Top Layer - Bottom Layer	Pad	Rounded				
	2	3.200mm(125.984mil)	Non-Plated	Round	Top Layer - Bottom Layer	Pad	Rounded				
	8	3.500mm(137.795mil)	Non-Plated	Round	Top Layer - Bottom Layer	Pad	Rounded				
	2	10.000mm(393.701mil)	Non-Plated	Round	Top Layer - Bottom Layer	Pad	Rounded				
	414 Total										

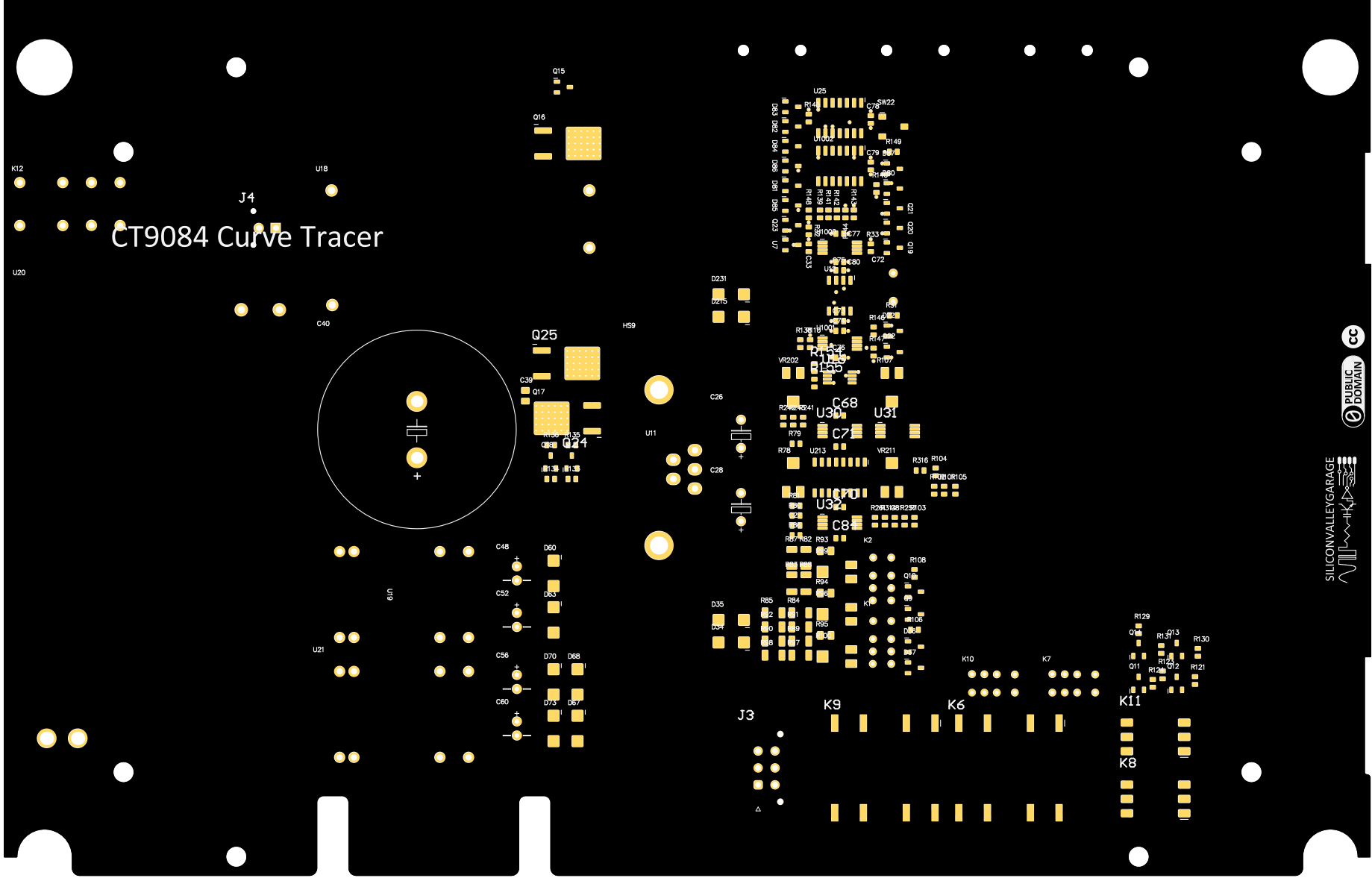
DRILL DRAWING

Drill Drawing View (Scale 1:1)

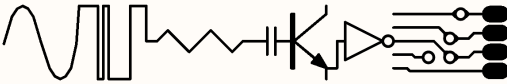


COMPOSITE VIEW FRONT

Top Layer (Scale 1:1)



SILICONVALLEYGARAGE



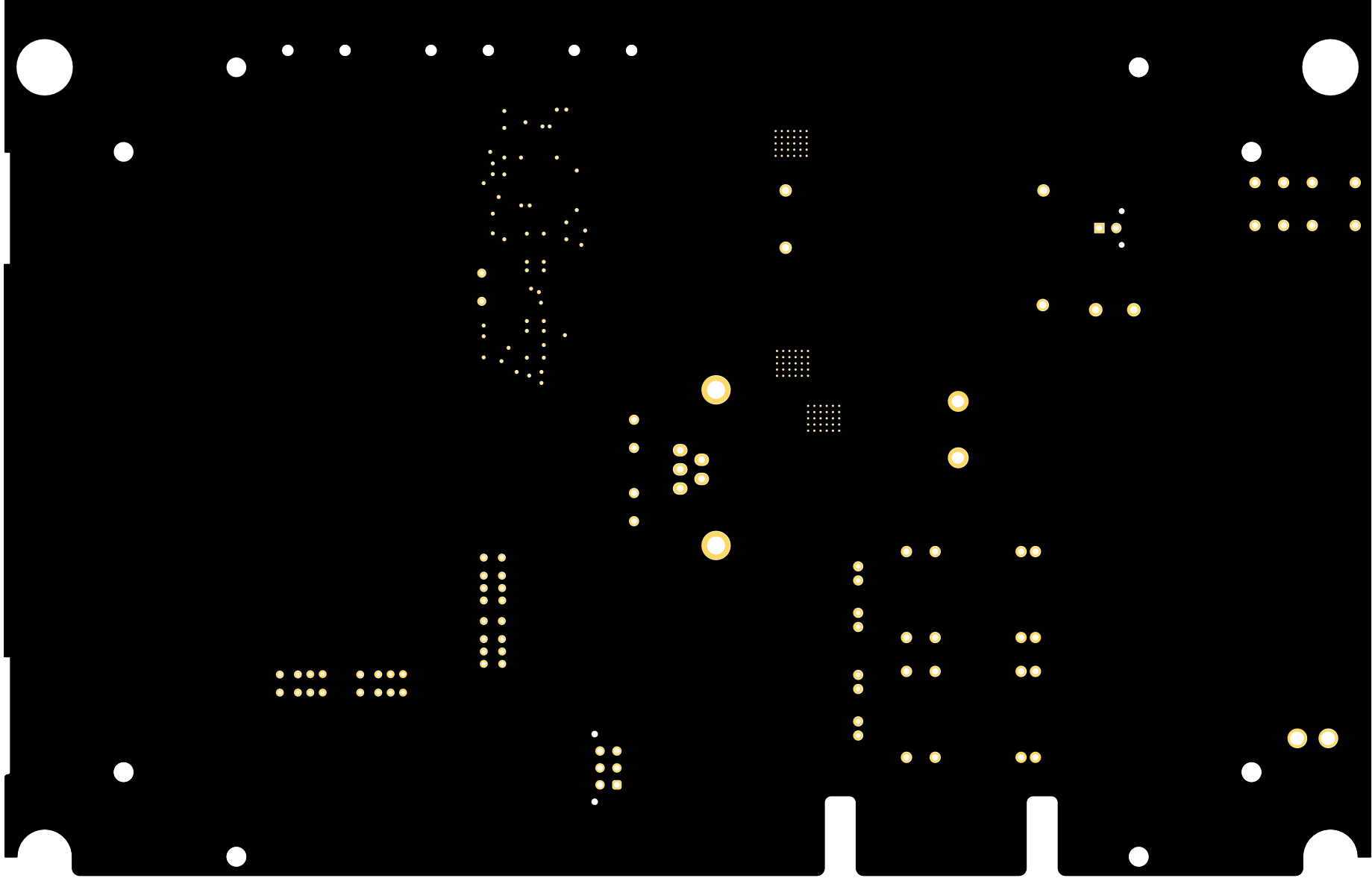
Project CT9042.PrjPcb

Version: | Variant [No Variations]

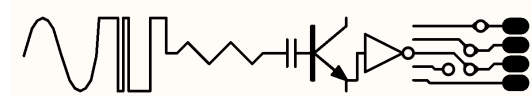
FABRICATION DRAWING

COMPOSITE VIEW BACK

Bottom Layer (Scale 1:1)



SILICONVALLEYGARAGE

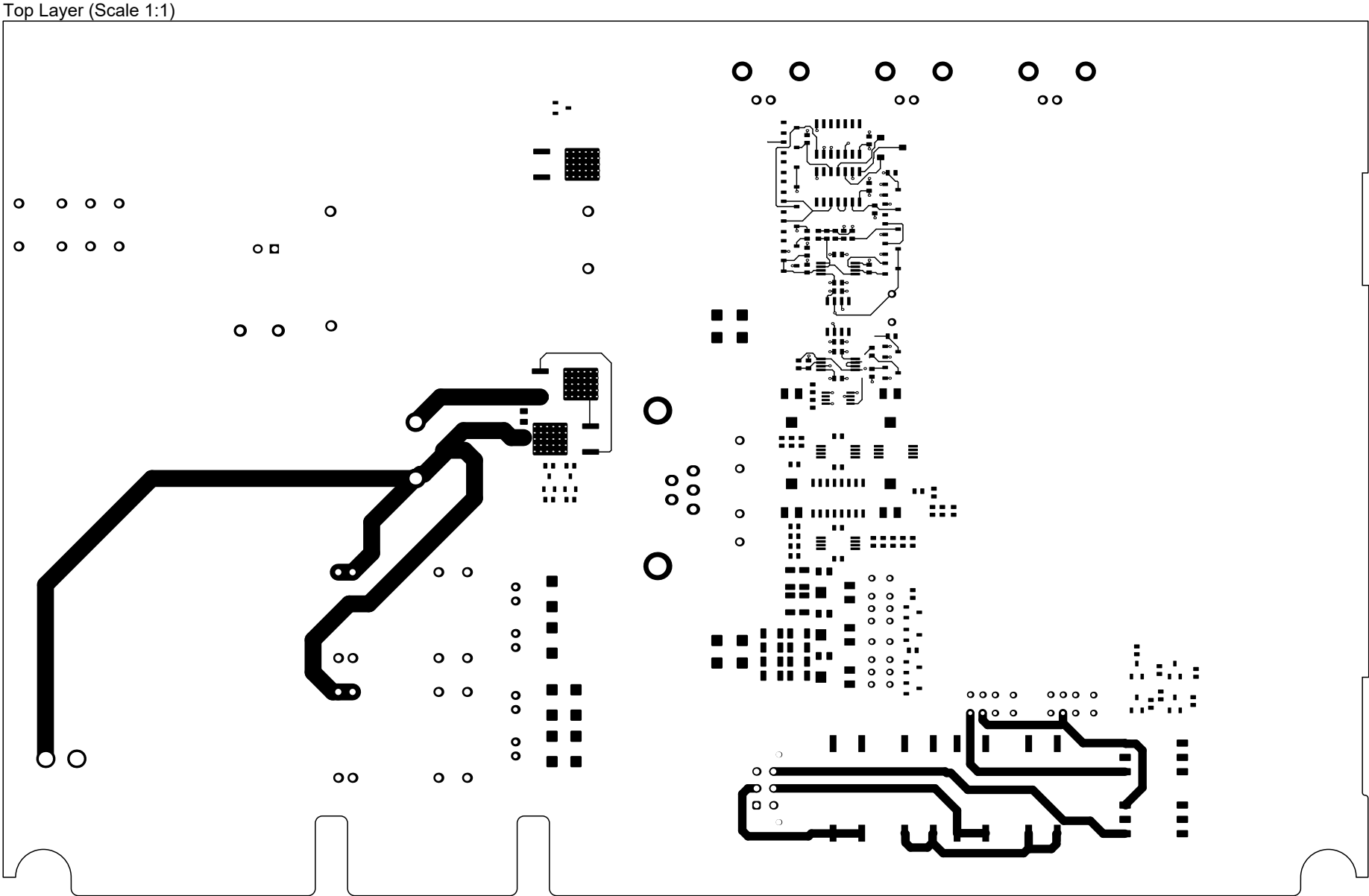


Project CT9042.PrjPcb

Version: | Variant [No Variations]

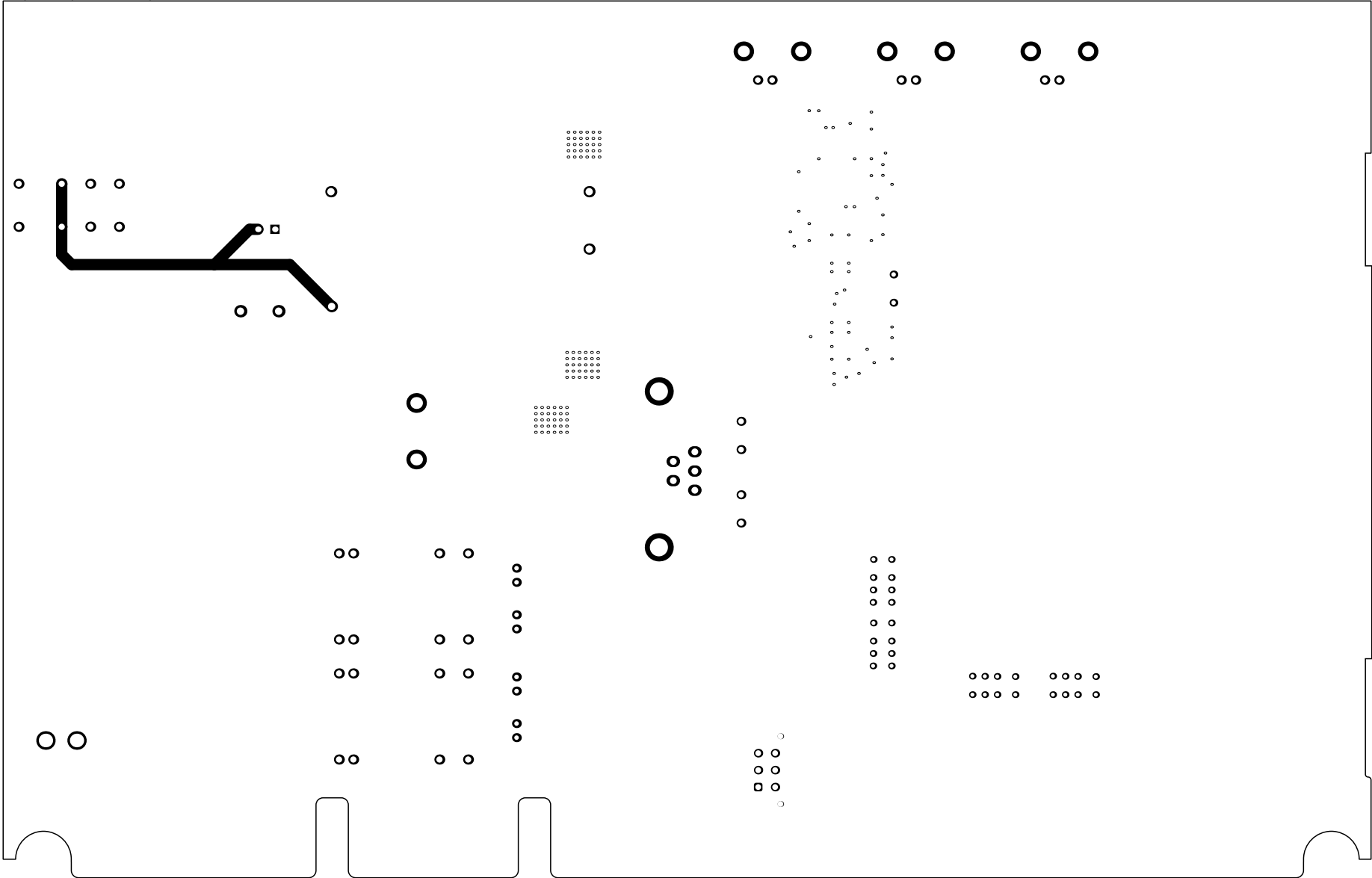
FABRICATION DRAWING

LAYER VIEW : TOP LAYER



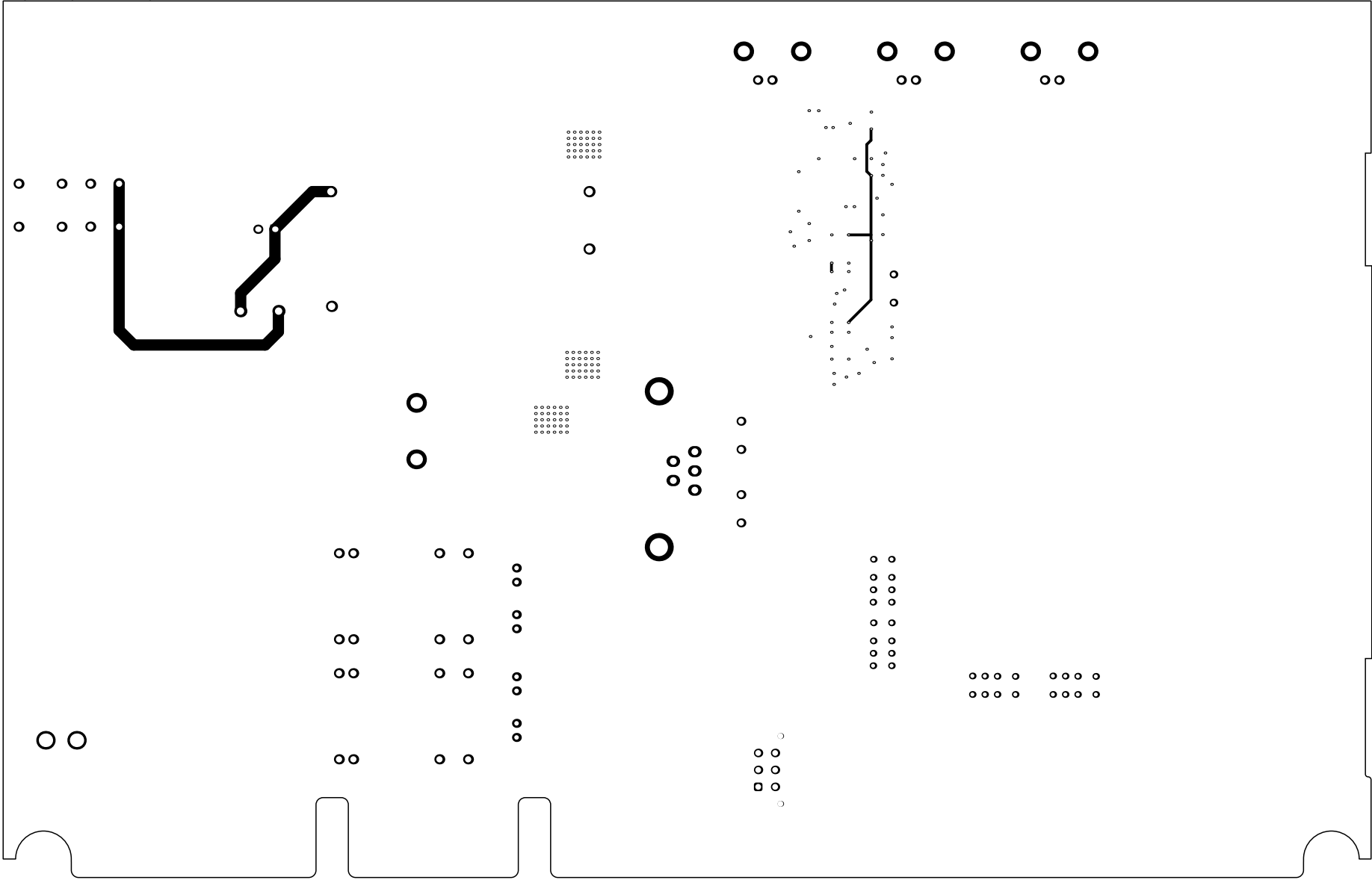
LAYER VIEW : MID LAYER 1

Layer 1 (Scale 1:1)



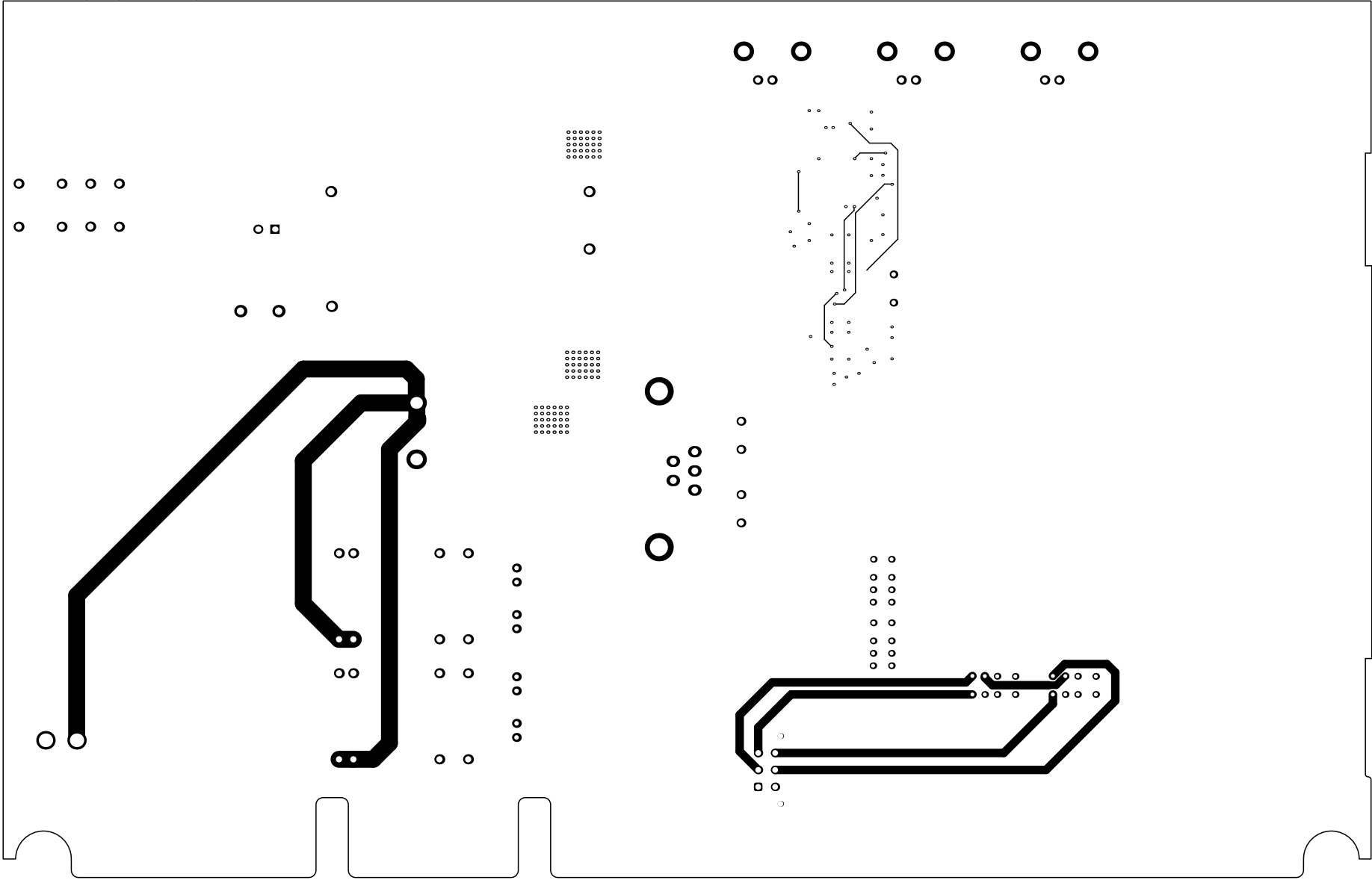
LAYER VIEW : MID LAYER 2

Layer 2 (Scale 1:1)



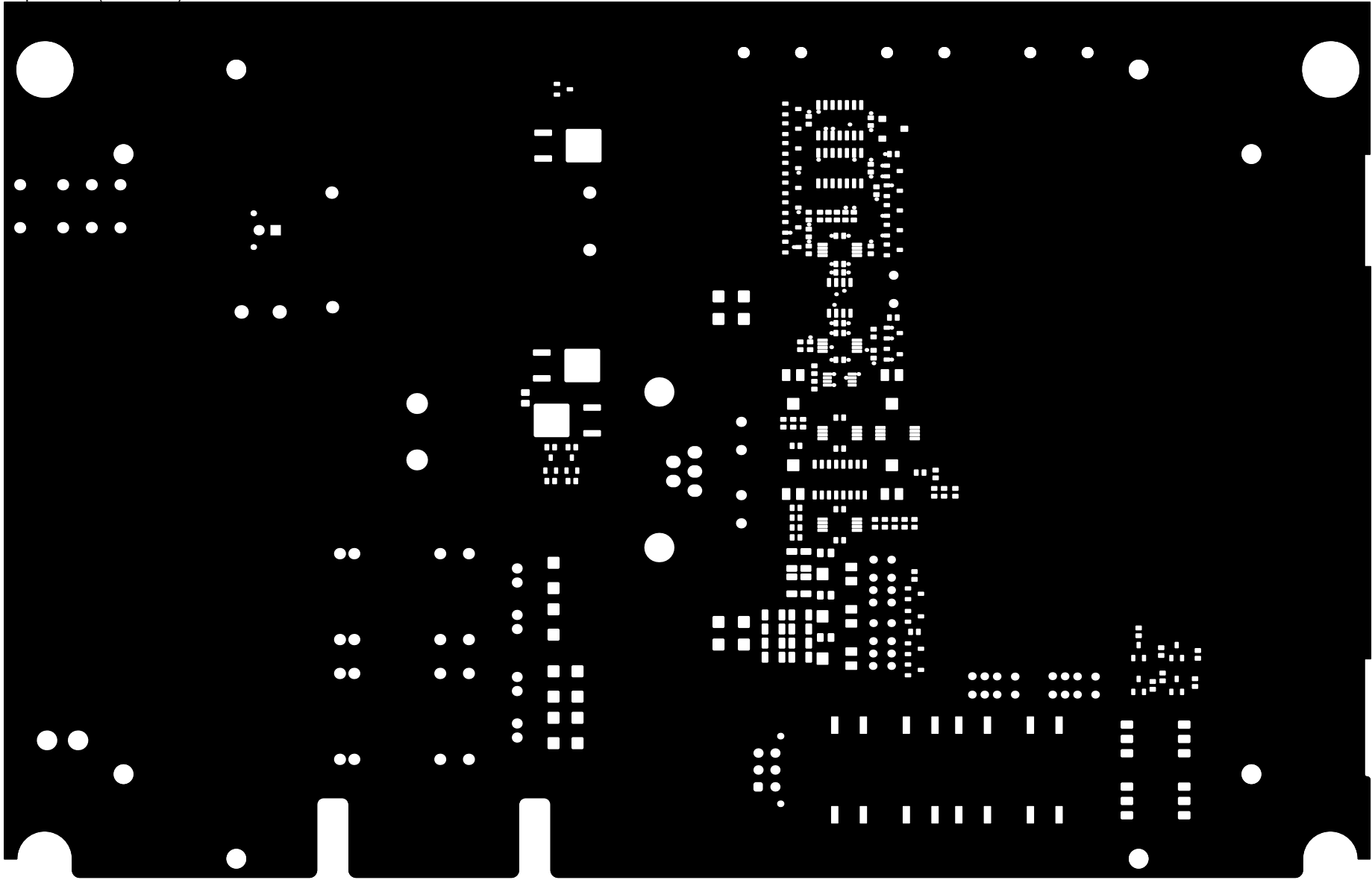
LAYER VIEW : BOTTOM LAYER

Bottom Layer (Scale 1:1)



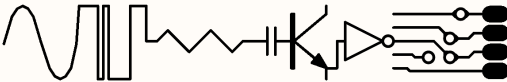
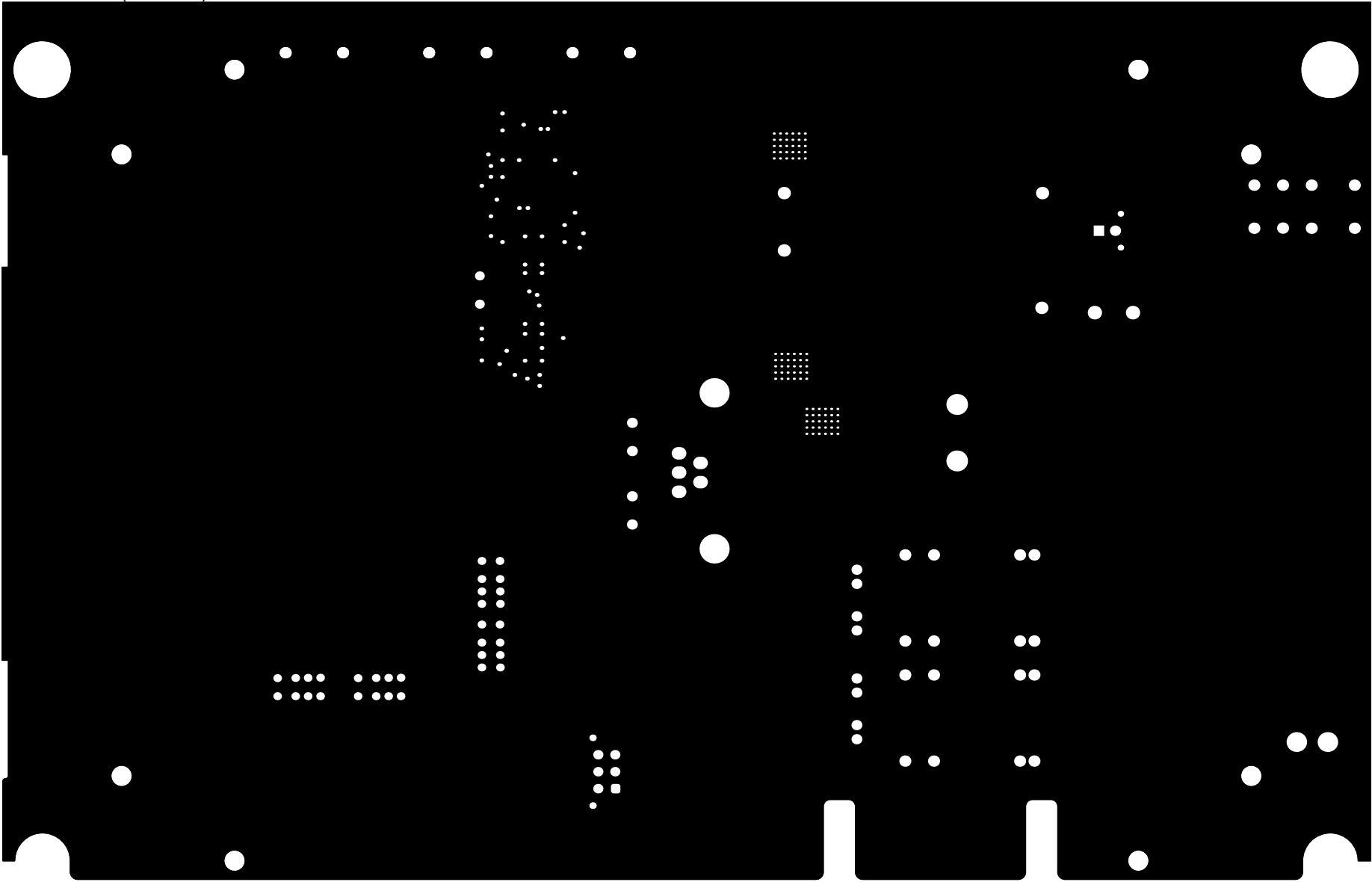
LAYER VIEW : TOP SOLDER MASK

Top Solder (Scale 1:1)



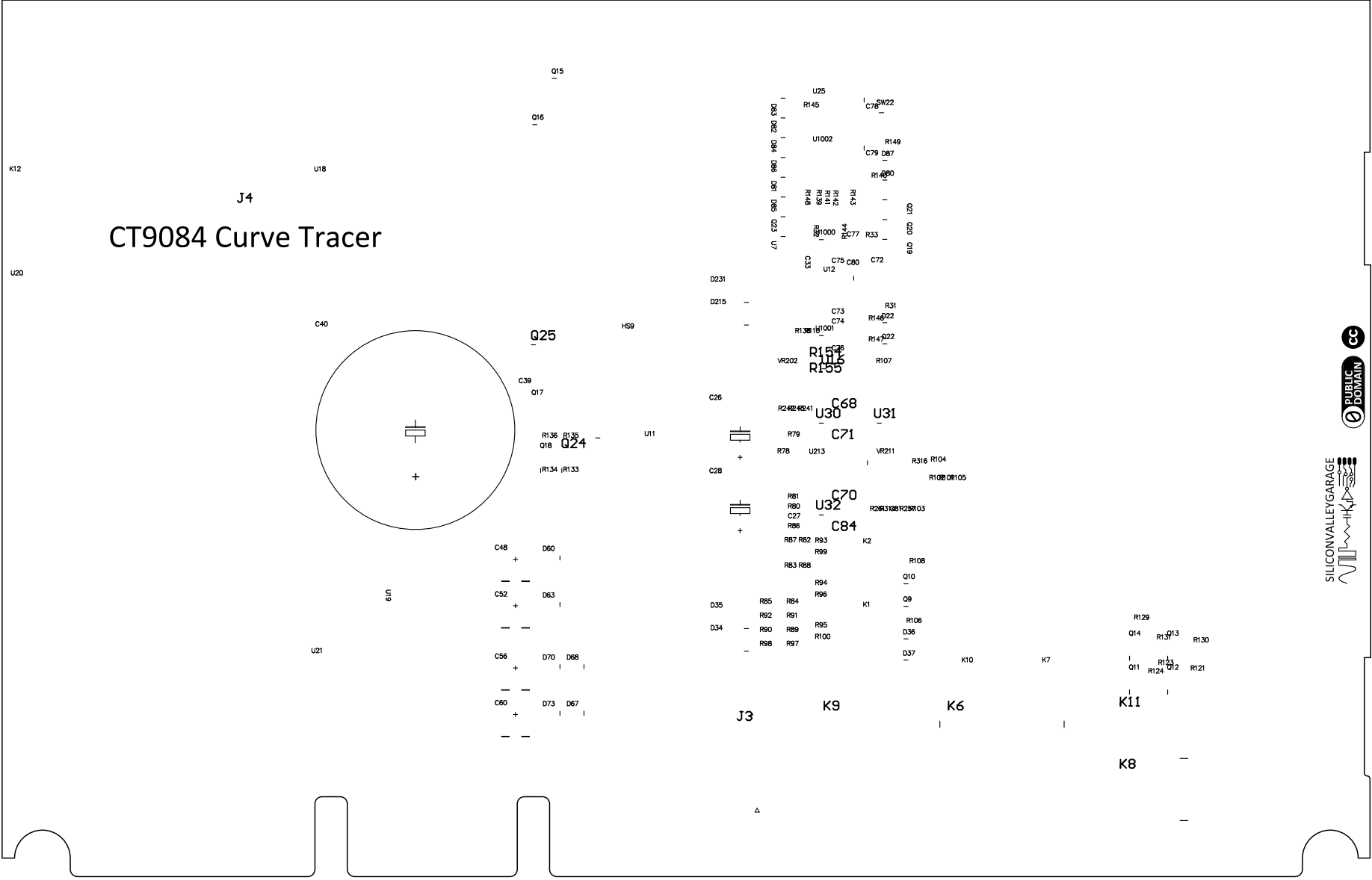
LAYER VIEW : BOTTOM SOLDER MASK

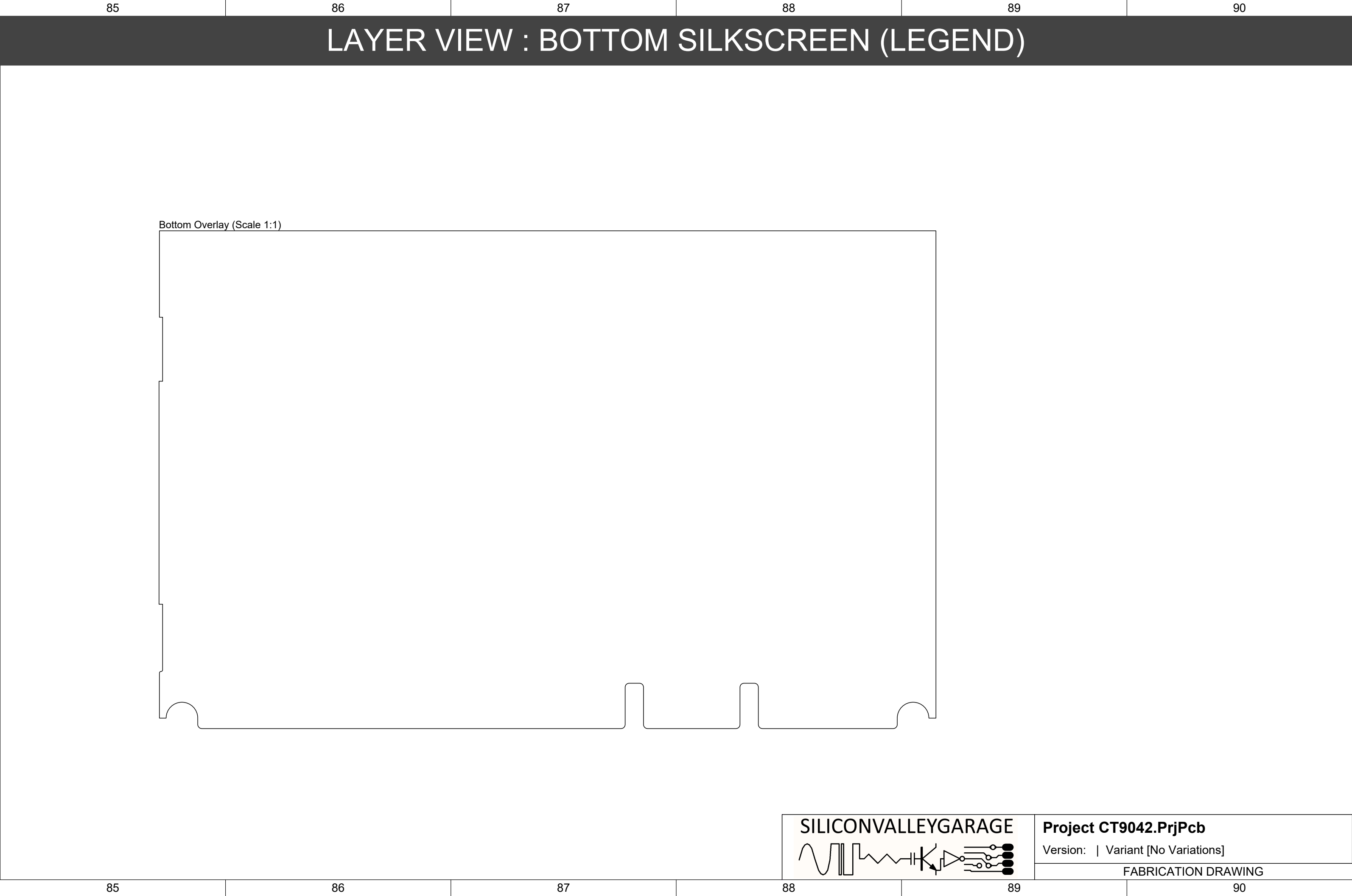
Bottom Solder (Scale 1:1)



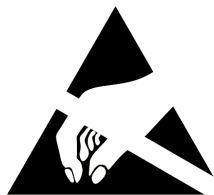
LAYER VIEW : TOP SILKSCREEN (LEGEND)

Top Overlay (Scale 1:1)





GENERAL



Unless otherwise specified the following rules apply:

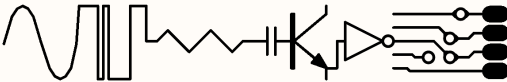
- 1. DO NOT DEVIATE FROM ARTWORK OR BOM WITHOUT PRIOR AUTHORIZATION.
- 2. ASSEMBLE AND INSPECT PER IPC-610 CLASS 2

Bill of Materials and Material Handling

- 3. THE BOM CONTAINED IN THIS DOCUMENT IS AS-BUILT. NON-INSTALLED PARTS HAVE BEEN REMOVED. ADDITIONAL BOM FORMATS ARE AVAILABLE IN THE PROJECT FILES
- 4. ANY PART SUBSTITUTIONS MUST BE APPROVED IN WRITING BEFORE ASSEMBLY
- 5. ALL MATERIALS MUST BE PROCURED FROM MANUFACTURER AUTHORIZED DISTRIBUTORS OR THE ORIGINAL MANUFACTURER
- 6. ALL COMPONENTS AND BOARDS TO BE HANDLED AND STORED ACCORDING TO IPC GUIDELINES
- 7. ESD CONTROL PER IPC RULES

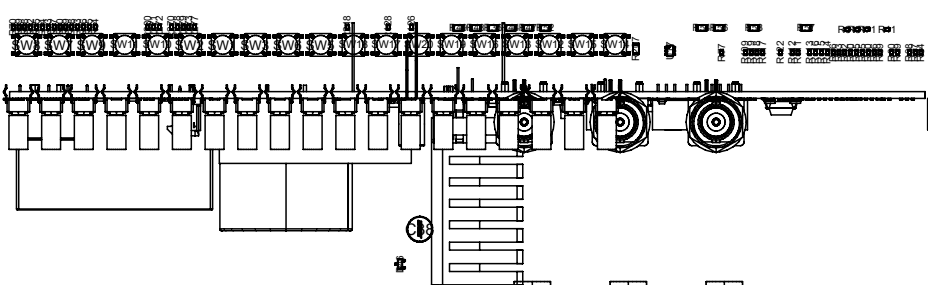
Soldering

- 8. SOLDERING TO BE DONE USING SN37PB63 ALLOY USING ALLOY MANUFACTURER RECOMMENDED NO-CLEAN FLUX
- 9. BGA COMPONENTS WITH LEAD-FREE CONNECTIONS NEED TO BE REBALLED WITH SN63PB37. MIXING OF ALLOYS IS NOT PERMITTED.
- 10. SOLDERING PREFERRABLY TO BE DONE USING NITROGEN ATMOSPHERE
- 11. SURPLUS COMPONENTS TO VACUUM SEALED WITH DESSICANT IN ANTISTATIC BAGS
- 12. INCOMING MATERIAL (BOARDS AND COMPONENTS) NEEDS TO BE INSPECTED FOR HUMIDITY AND BAKED IF NEEDED PRIOR TO USE.
- 13. MANUAL REWORK / TOUCHUP TO BE DONE USING SAME ALLOY AND APPROPRIATE FLUX. FLUX MUST BE REMOVED.

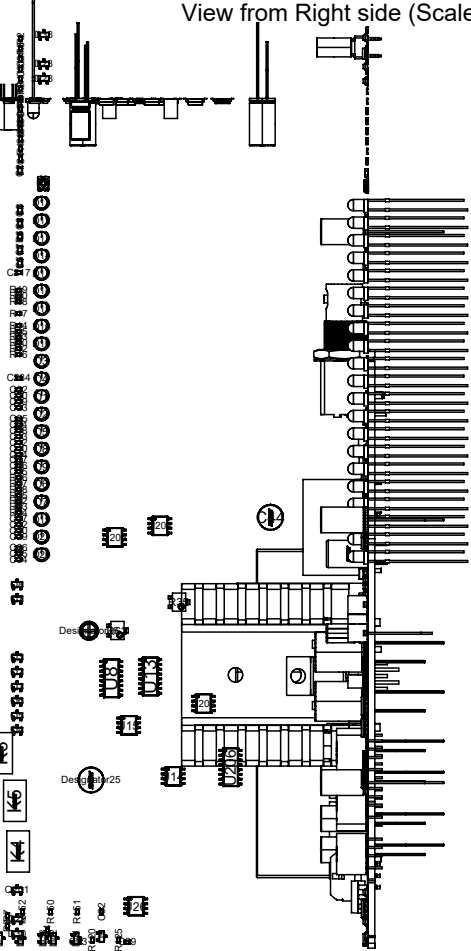


2D VIEW

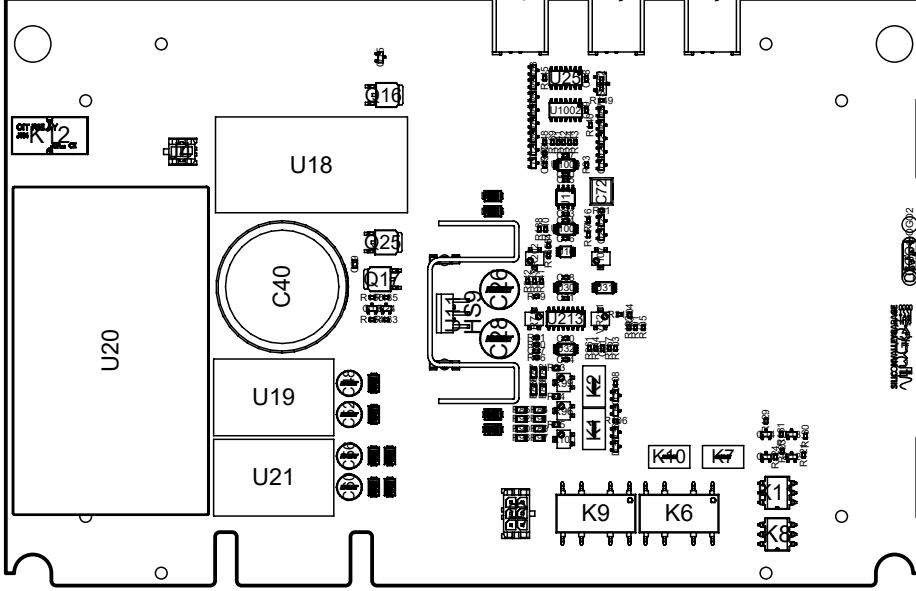
View from Back side (Scale 1:2)



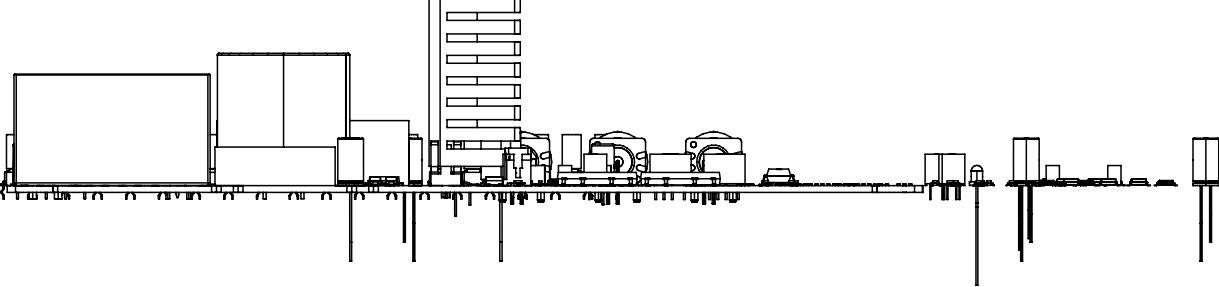
View from Right side (Scale 1:2)



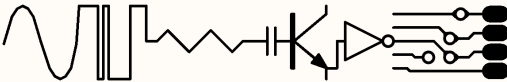
View from Top side (Scale 1:2)



View from Front side (Scale 1:2)



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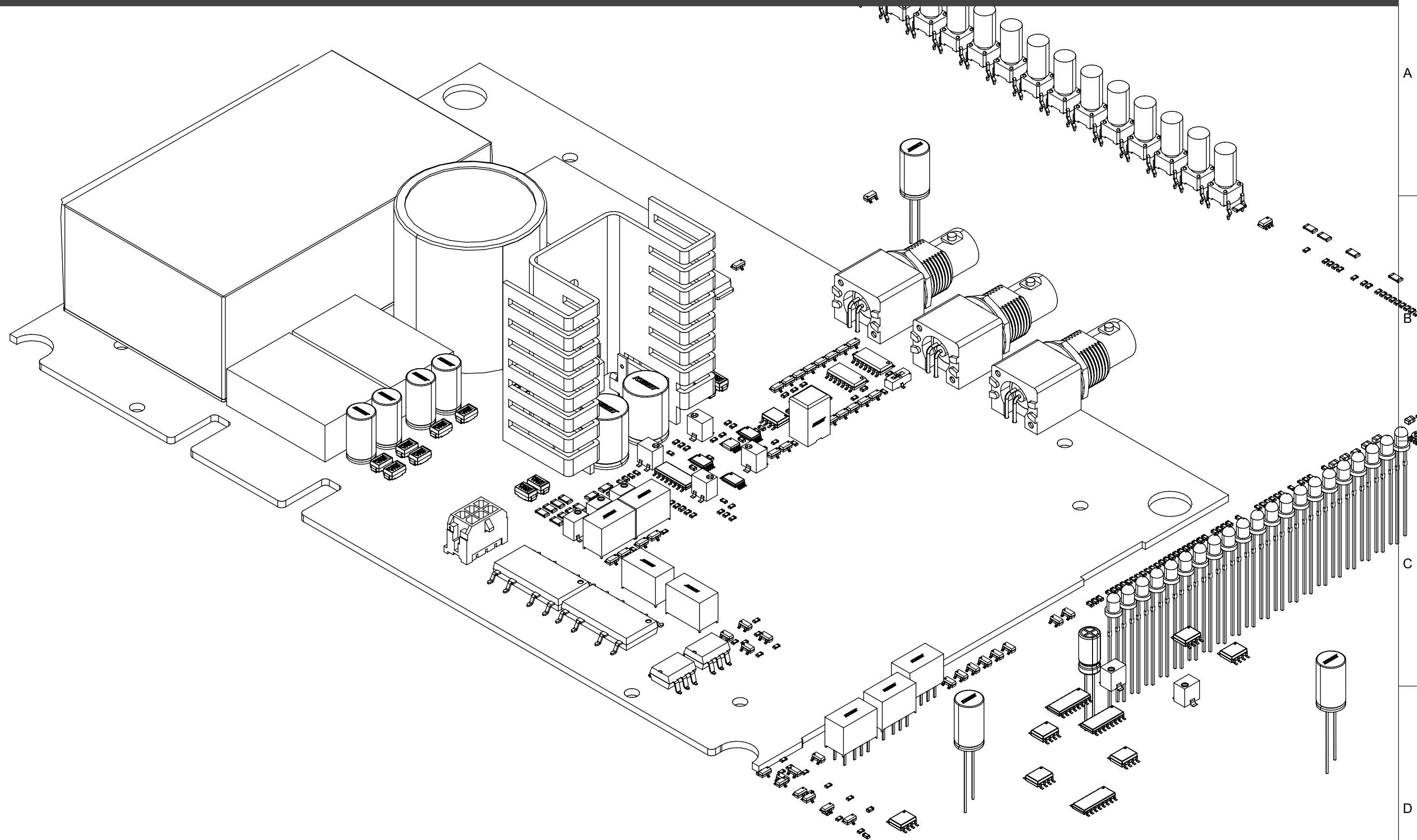


Project CT9042.PrjPcb

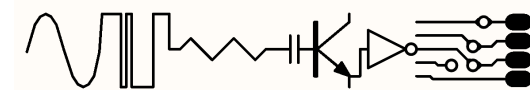
Version: | Variant [No Variations]

ASSEMBLY DRAWING

3D VIEW



SILICONVALLEYGARAGE



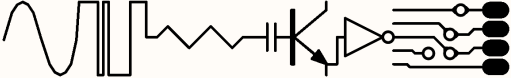
Project CT9042.PrjPcb

Version: | Variant [No Variations]

ASSEMBLY DRAWING

Bill Of Materials

Quantity	Designator	Description	LCSC	MOUSER
40	C1, C4, C6, C8, C10, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C22, C24, C25, C29, C30, C31, C32, C36, C37, C68, C69, C70, C71, C73, C74, C75, C76, C77, C78, C79, C80, C83, C84, C?	CAPACITOR,CERAMIC,100nF,50V,X7R,0603	C127833	80-C0603C104K5R
5	C2, C3, C5, C7, C9	CAPACITOR,CERAMIC,10uF,25V,X5R,0805	C3039694	187-CL21B106KAYQNNE
2	C21, C33	CAPACITOR,CERAMIC,10nF,50V,X7R,0603		
3	C23, C34, C217	CAPACITOR,CERAMIC,100pF,50V,C0G,0603		
2	C26, C28	CAPACITOR,ALU,47uF,100V,TH,P5mm,D10mm,H14mm		
2	C27, C234	CAPACITOR,CERAMIC,10pF,50V,C0G,0603		80-C0603C100J5GAC
2	C35, C47	CAPACITOR,CERAMIC,1nF,50V,C0G,0603		
6	C38, C44, C48, C52, C56, C60	CAPACITOR,ALU,47uF,35V,TH,P0.25mm,D6.3mm,H12.5mm		
1	C39	CAPACITOR,CERAMIC,100nF,100V,X7R,0805		
1	C40	CAPACITOR,ALU,2200uF,100V,TH,P10mm,D35mm,H32mm,SNAP		
4	C41, C43, C45, C46	CAPACITOR,CERAMIC,10nF,100V,C0G,0805		
1	C42	CAPACITOR,CERAMIC,4u7,10V,X5R,0603	C1705	187-CL10A475KP8NNNC
1	C72	CAPACITOR,POLYPROP,100nF,100V		
1	C81	CAPACITOR,CERAMIC,1n5,50V,X7R,0603		
21	D1, D2, D3, D4, D5, D6, D7, D8, D9, D10, D11, D12, D13, D14, D15, D16, D17, D18, D19, D20, D21	LED,TH,GREEN,42mCd,3MM		78-VLMPG33N1P2
7	D22, D23, D28, D29, D30, D38, D87	DIODE,75V,250mA,SOT23,500pA ULTRA LOW LEAKAGE		
8	D24, D25, D26, D27, D39, D43, D44, D45	DIODE,RECTIFIER,75V,0.3A,MMBD4148,SOT23		
17	D31, D32, D33, D36, D37, D41, D42, D46, D80, D81, D82, D83, D84, D85, D86, D?	DIODE,RECTIFIER,75V,0.3A,BAS16,SOT23		
10	D34, D35, D60, D63, D67, D68, D70, D73, D215, D231	DIODE,RECTIFIER,1000V,1A,ONSEMI,NRVA4007,SMA		
1	DISP1	DISPLAY,LCD,ALPHA,2X40CHAR,HD44780,YELLOW BL		
1	HS9	HEATSINKSTAMPED,CLIP-ON,BOARDMOUNT,3W,25C/W,CUI,HSS01-B20-CP		
2	J1, J2			
1	J3	CONN,2X3,MOLEX MICROFIT,TH,VERT		
1	J4	CONN,1X2,MOLEX, MICROFIT,VERT		
1	J9	CONN,POWER,IEC,SWITCH,FUSE ASSEMBLY,PANELMOUNT		
1	JX			
1	JY			
1	JZ			
7	K1, K2, K3, K4, K5, K7, K10	RELAY,DPDT,1A,250VAC,220VDC,5V COIL,KEMET,UA2-5NJ		
2	K6, K9	RELAY,OPTOMOS,6A,60V,AC,IXYS,CPC1907B,SMD		
2	K8, K11	RELAY,OPTOMOS,1.5A,100V,AC,IXYS,LCA701S,SMD		
1	K12	RELAY,DPDT,5V COIL,250VAC,3A		
13	Q1, Q2, Q4, Q6, Q7, Q8, Q9, Q10, Q11, Q12, Q13, Q14, Q24	XSTR,NPN,45V,0.1A,,BC847B,SOT23		
5	Q3, Q5, Q18, Q19, Q201	XSTR,PNP,SOT23,BC857,SOT23		
5	Q15, Q20, Q21, Q22, Q23	XSTR,NMOS,30V,4.3A,SI2336DS-T1-BE3.SOT23		
2	Q16, Q25	XSTR,PMOS,100V,5.6A,DMP10H400SK3-13,DPAK		
1	Q17	XSTR,NMOS,100V,12A,DMN10H170SK3-13,DPAK		
50	R1, R2, R3, R4, R29, R30, R39, R42, R51, R58, R60, R61, R67, R68, R70, R71, R76, R101, R102, R110, R120, R124, R125, R127, R131, R132, R133, R138, R139, R140, R141, R142, R144, R145, R146, R147, R148, R151, R152, R153, R154, R155, R156, R242, R243, R277, R279, R280, R?	RESISTOR,10K,0.1%,100mW,0603 (1608)		
1	R5	RES,TRIM,SMD,500R		652-TC33X-2-101E
1	R6	RESISTOR,GENERIC,0805		
27	R7, R8, R9, R10, R11, R12, R13, R14, R15, R16, R17, R18, R19, R20, R21, R22, R23, R24, R25, R26, R27, R28, R41, R119, R122, R126, R128	RESISTOR,511R,0.1%,100mW,0603 (1608)		

SILICONVALLEYGARAGE

Project CT9042.PrjPcb

Version: | Variant [No Variations]

ASSEMBLY DRAWING

